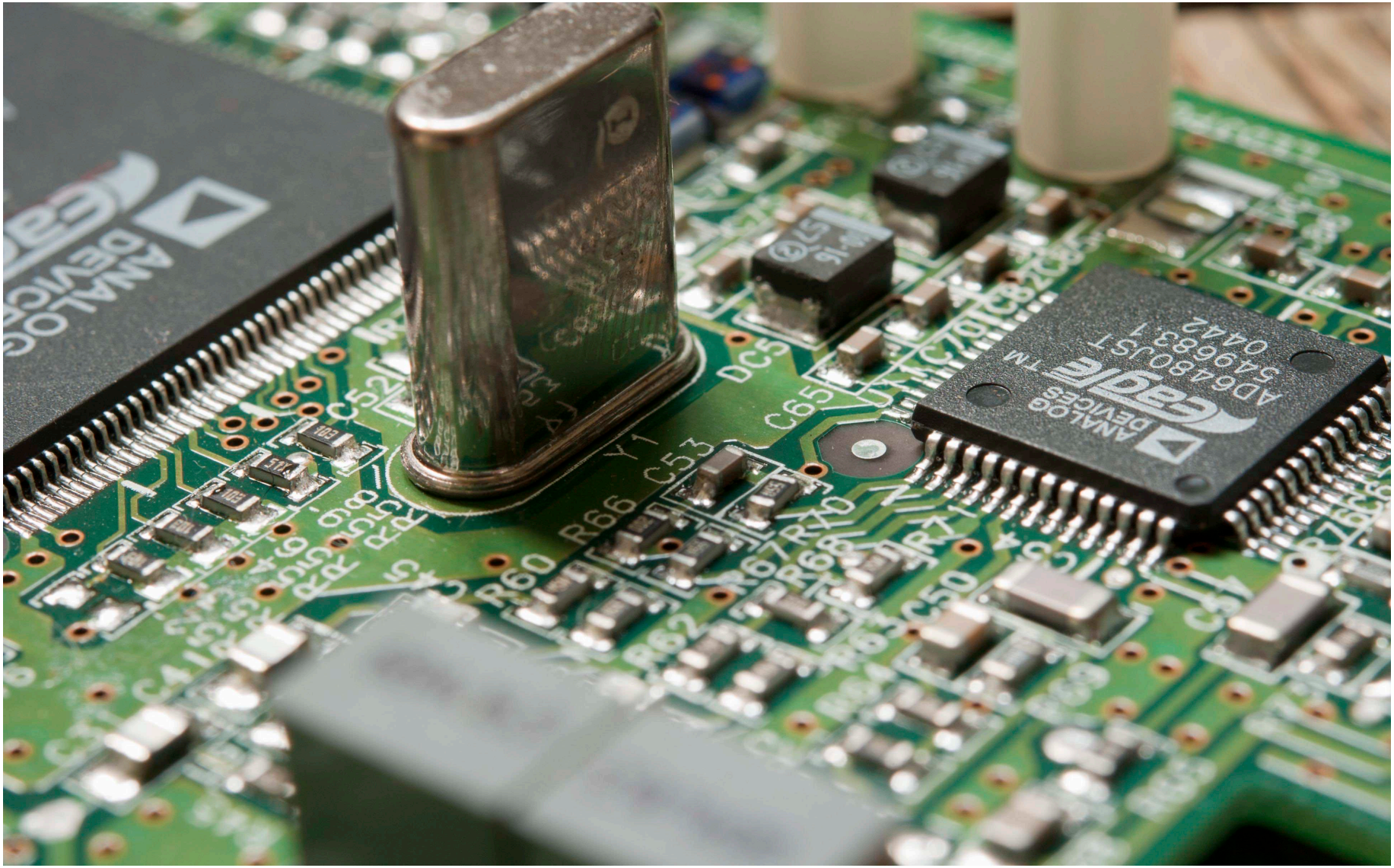
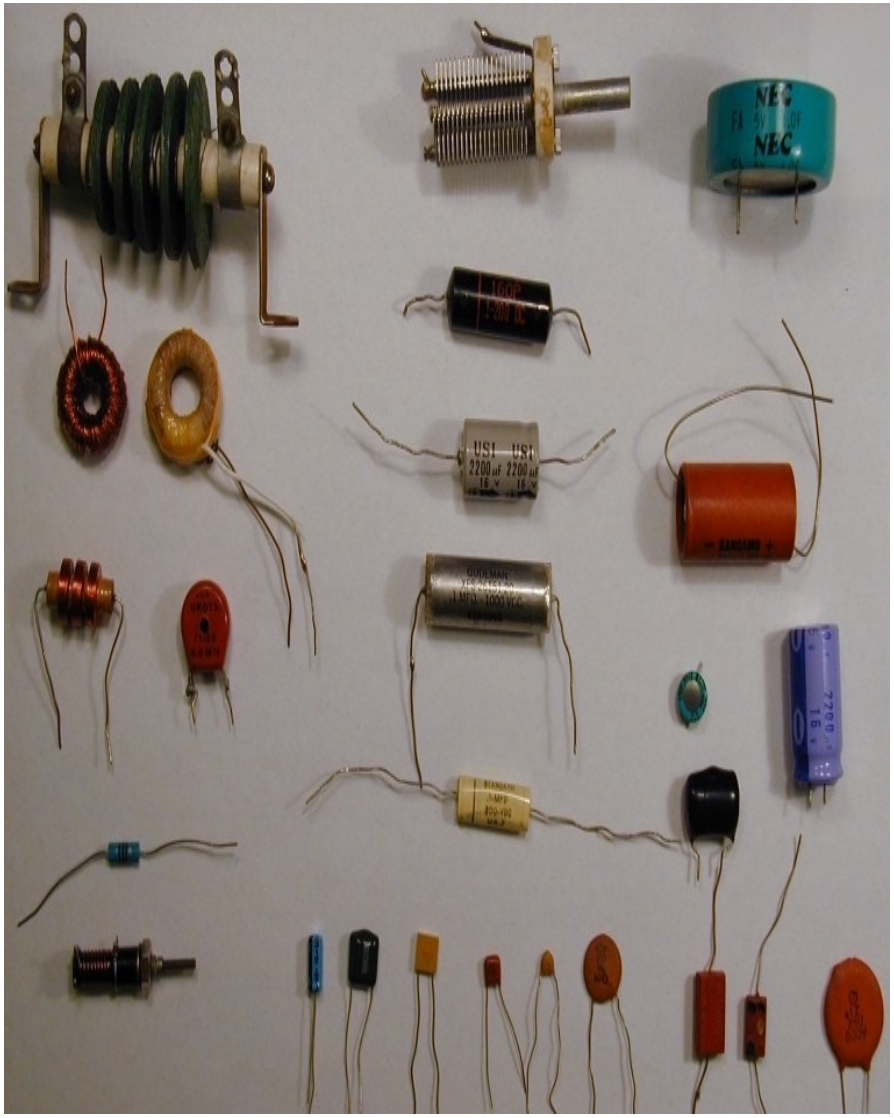
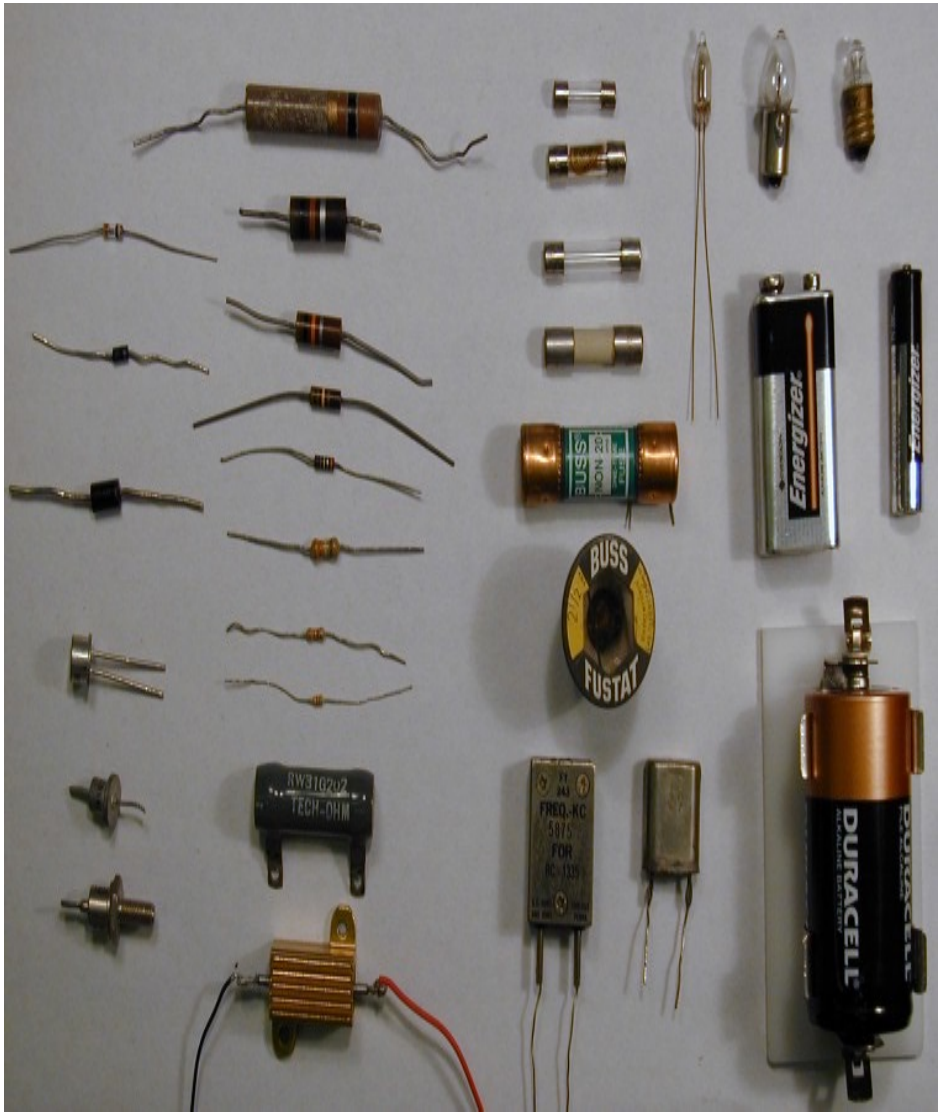
Introduction

It's All about the Transistor!



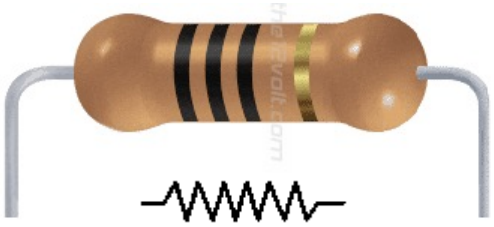
Passive Electronic Components





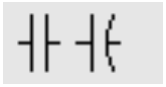
Passive Electronic Components

Resistors



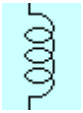
Capacitors



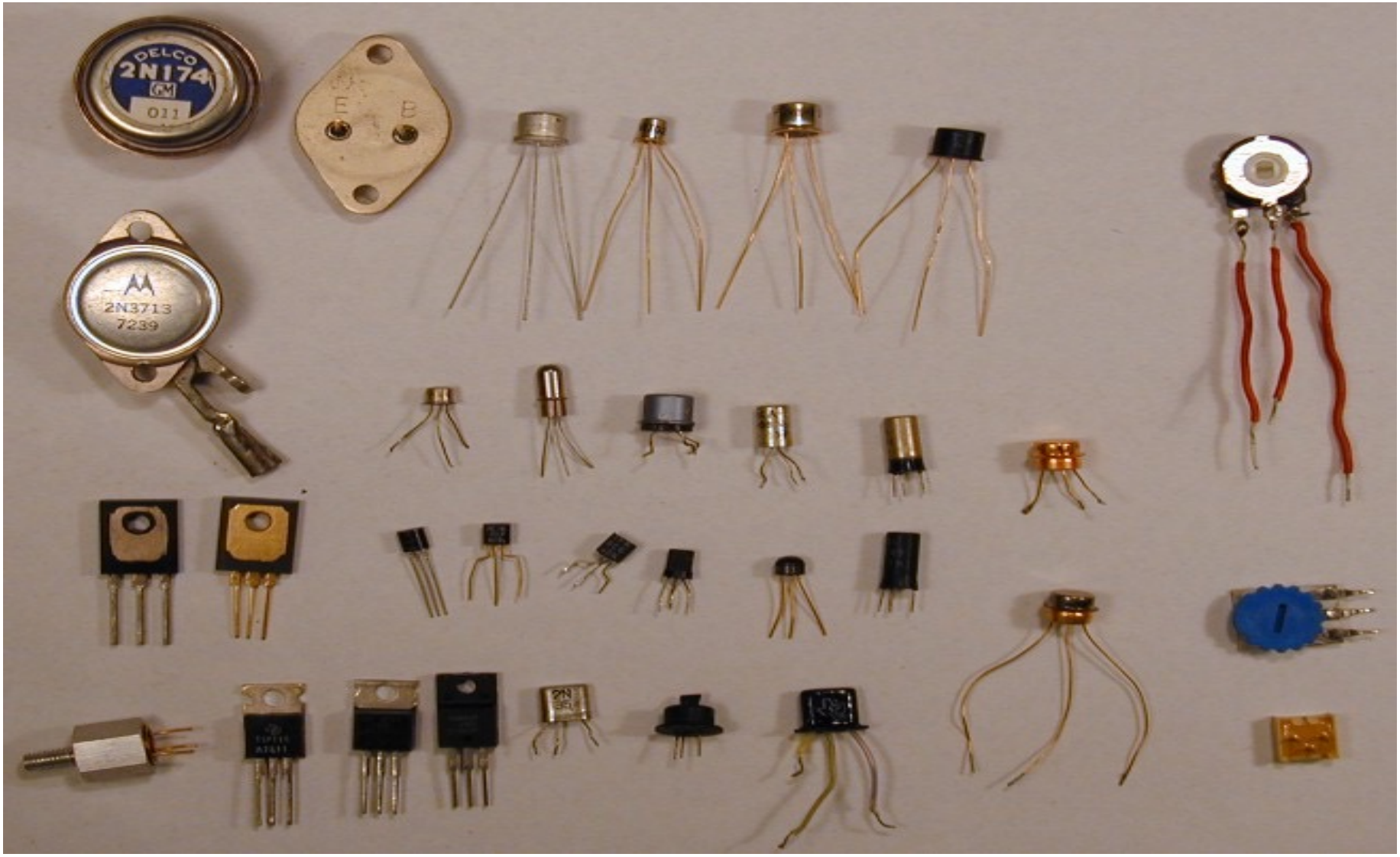


Inductors



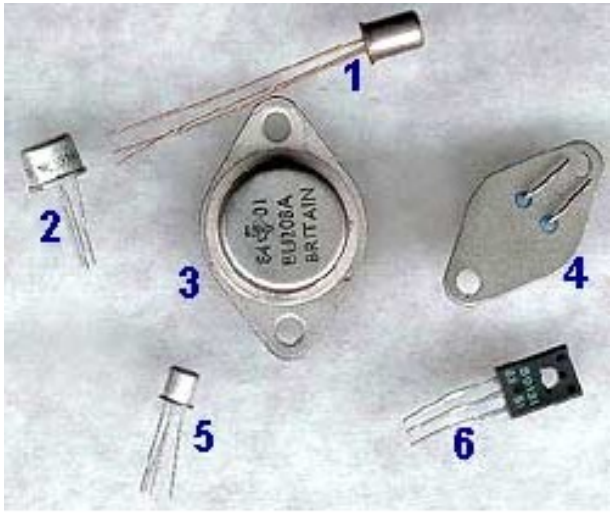


Active Electronic Components

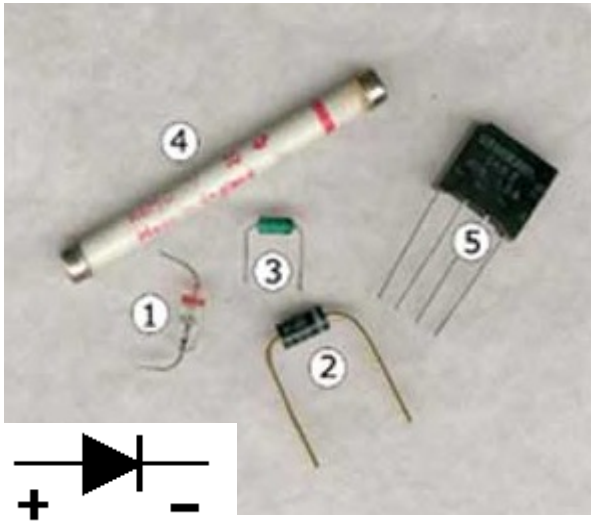


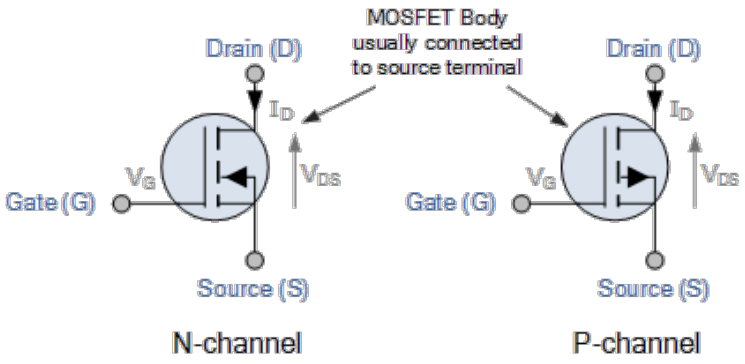
Active Electronic Components

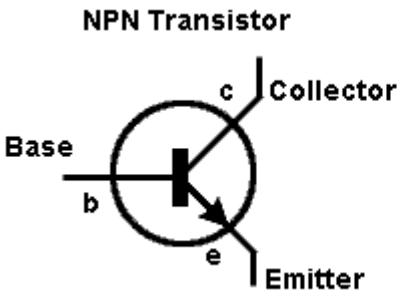
Transistors

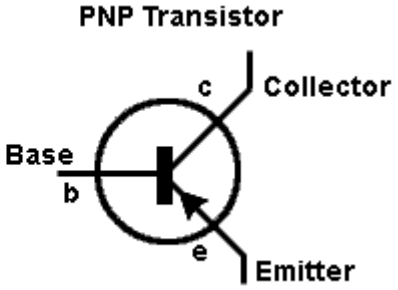


Diodes

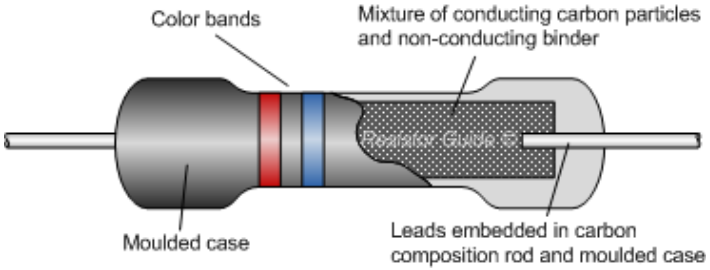


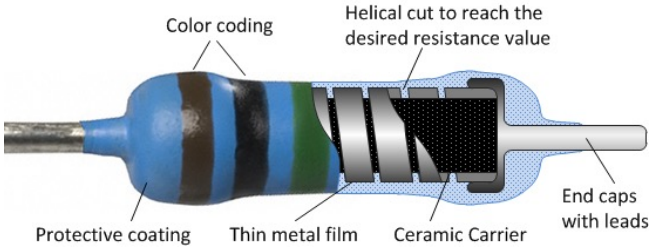


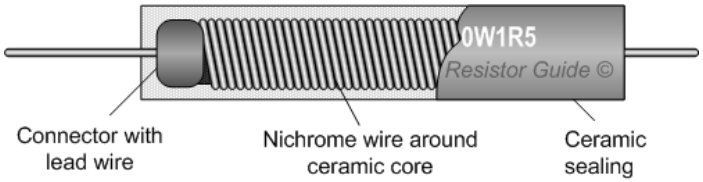


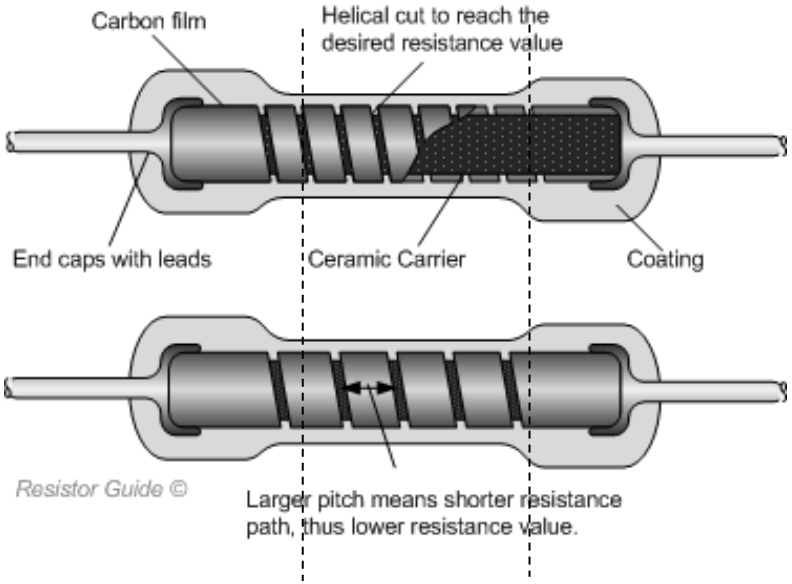


Passive Electronic Components
Resistors

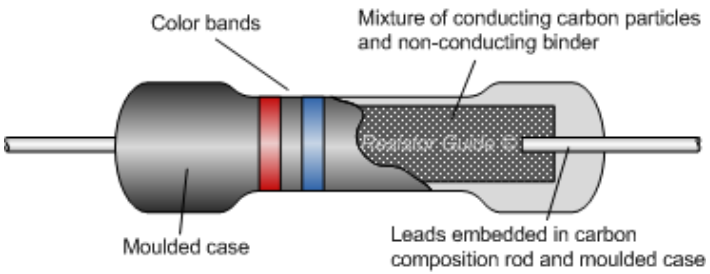


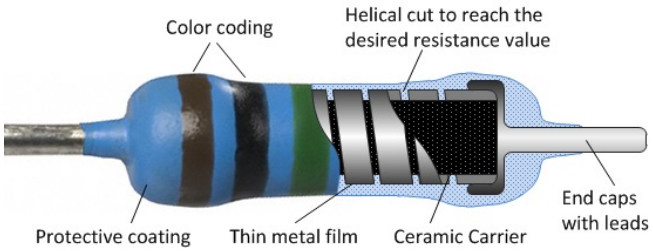






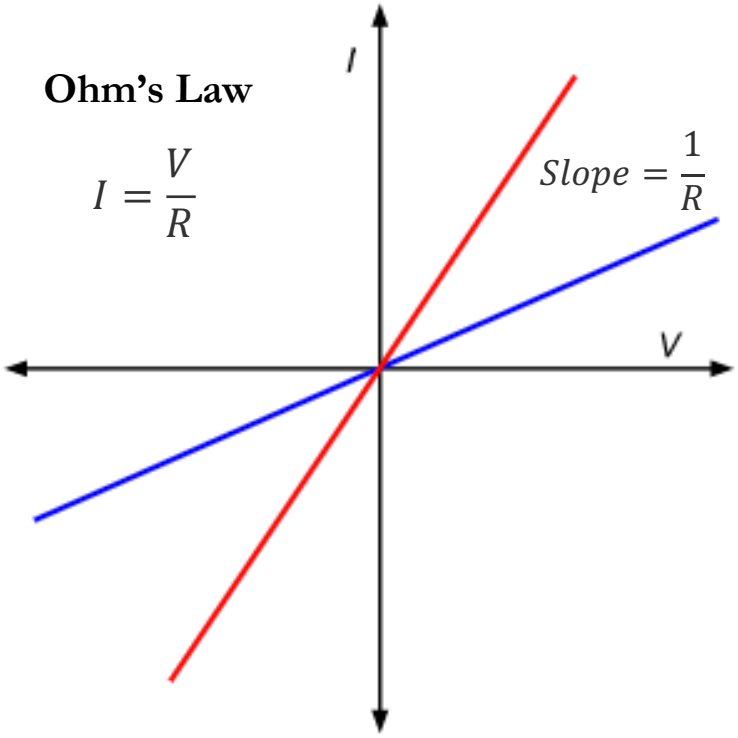
Passive Electronic Components
Resistors

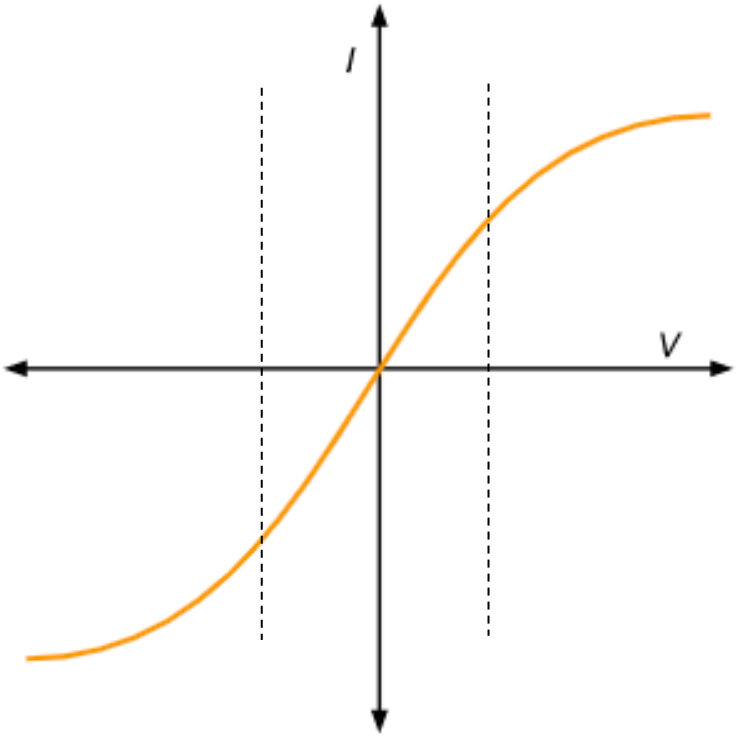




Ohm's Law

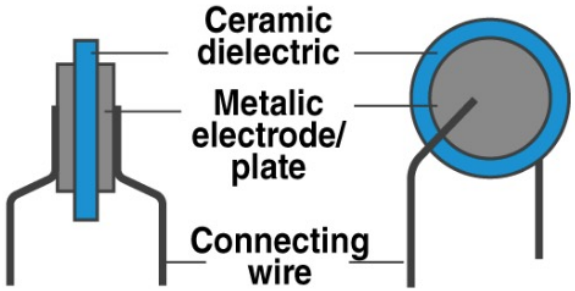
$$I = \frac{V}{R}$$

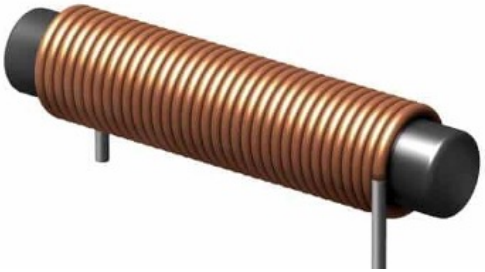




Passive Electronic Components
Capacitors and Inductors

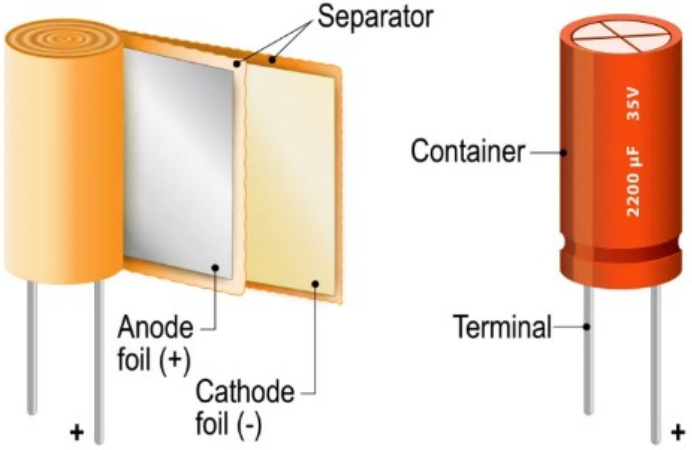


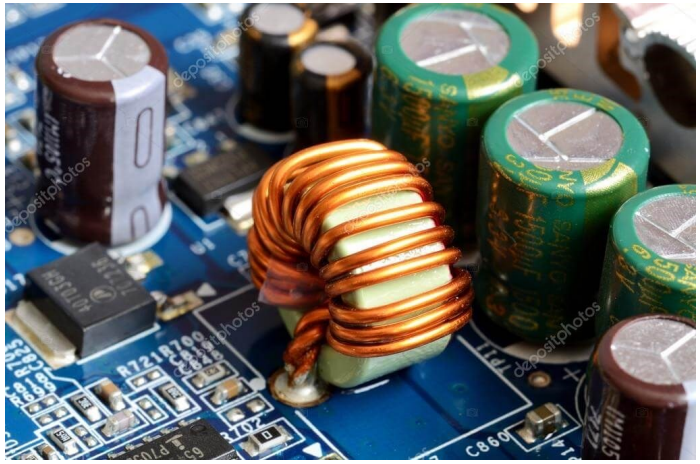




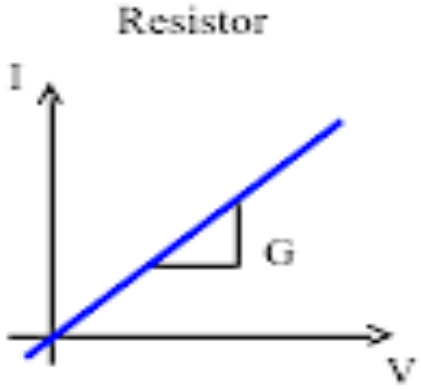




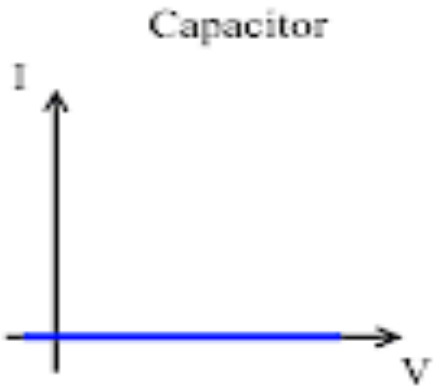




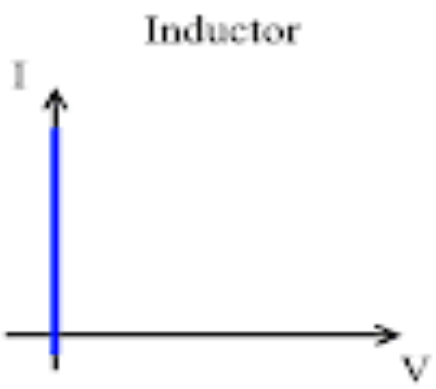
Passive Electronic Components
Capacitors and Inductors



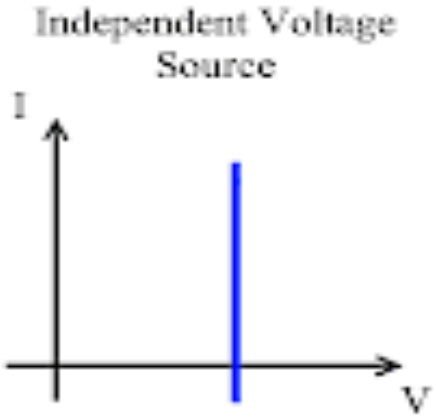
I = GV



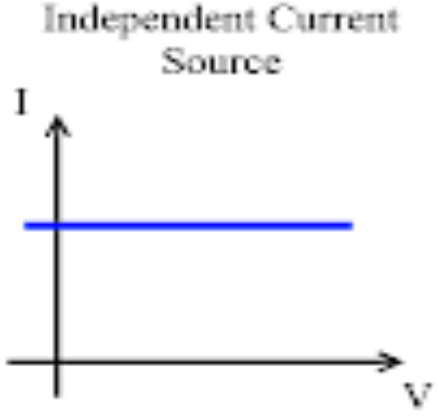
I = 0 for all V



V = 0 for all I



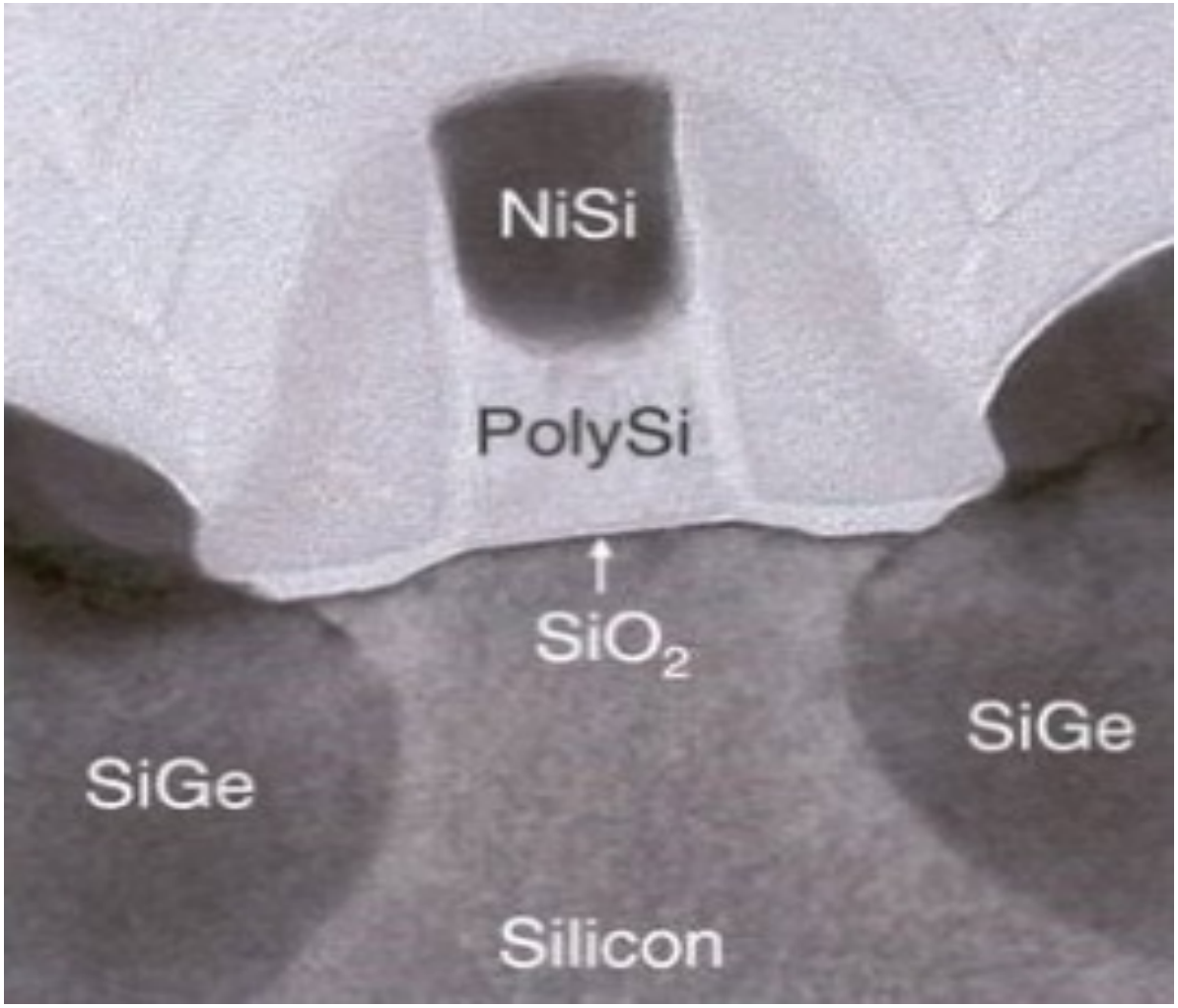
V constant for all I



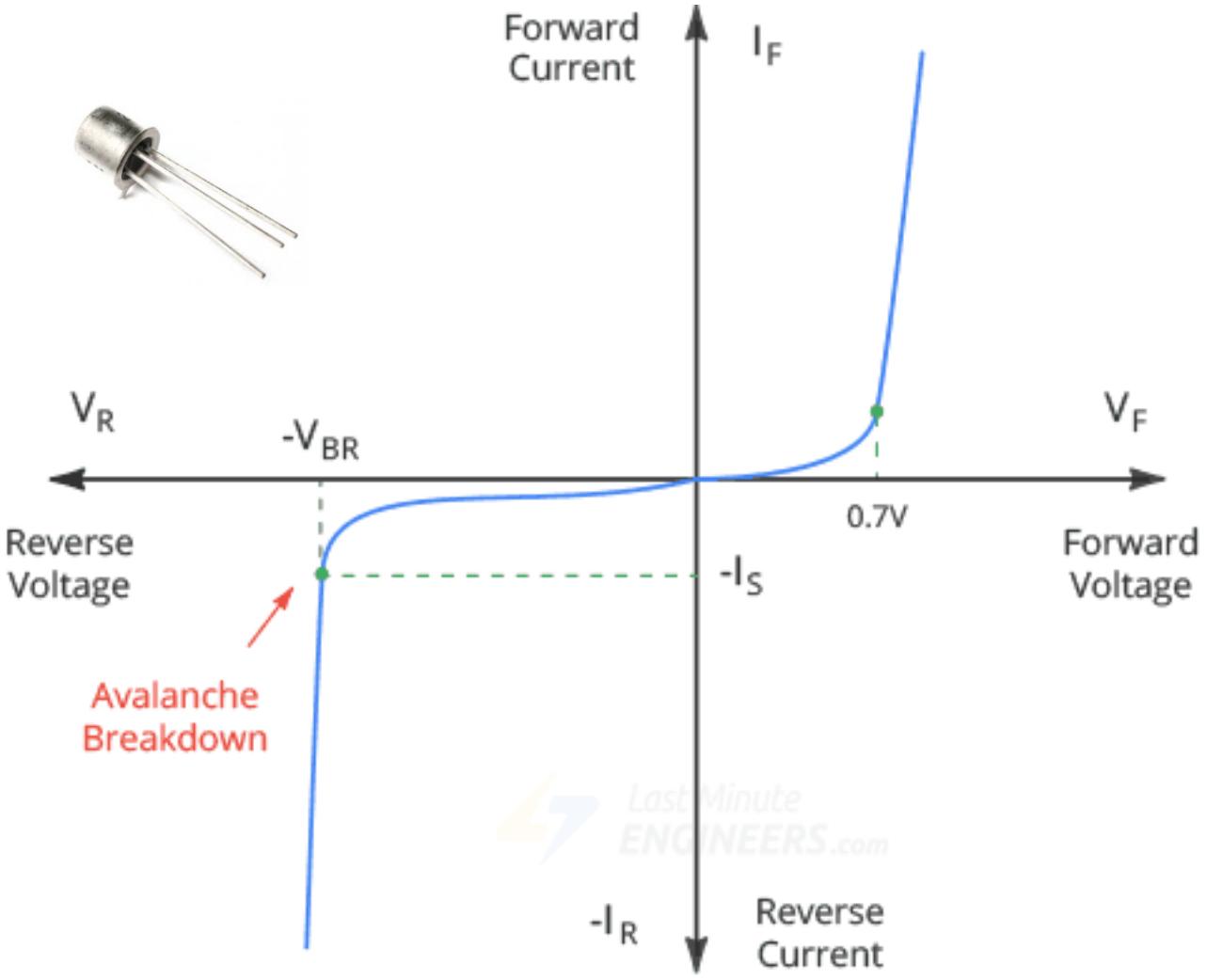
I constant for all V

Active Electronic Components

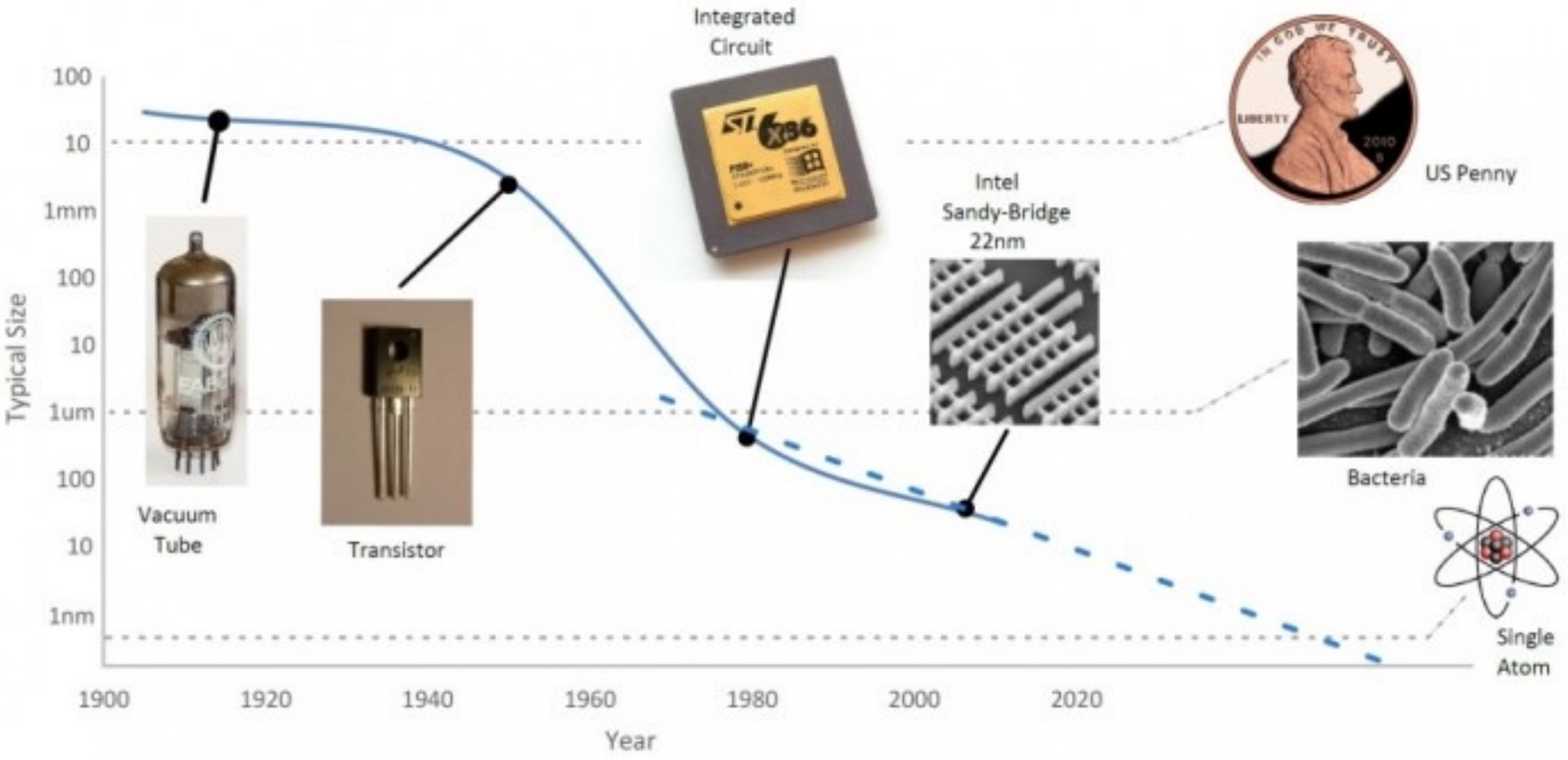




Active Electronic Components



It's All about the Transistor!



Competing Technologies

A successful technology should be able to implement two main processes.

1) Switching for computation.

2) Signal manipulation/processing.

The electronics technology has been so far remarkable in the implementation of both processes.

However, there are competing technologies in the making.

1) Electronics. Based on electron motion.

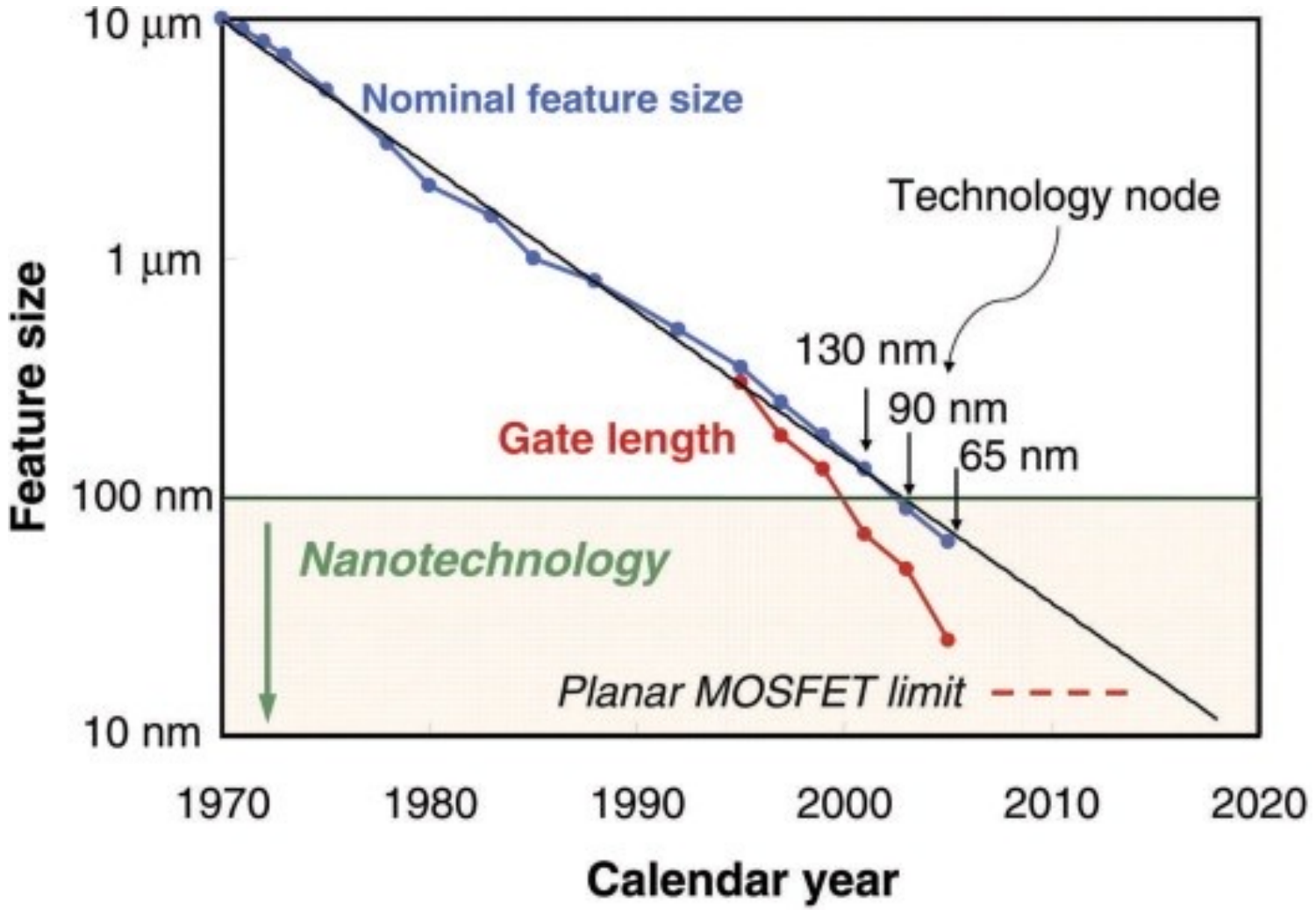
2) Photonics. Based on light wave (electronic transitions).

3) Quantum. Based on electron state.

It appears that electronics is quickly reaching its limits. WE NEED MORE!

The merge of infotech with biotech for our next transformation.

Transistor Evolution



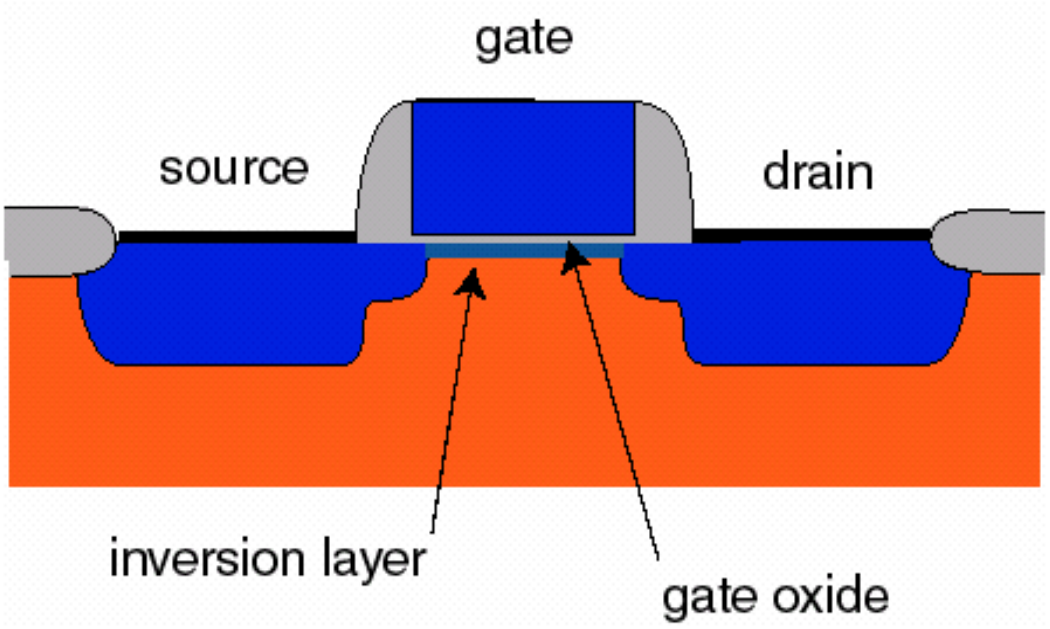
Keys to Success: Silicon



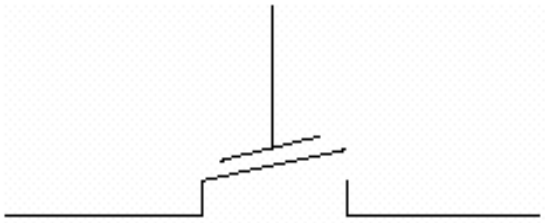
Keys to Success: The MOSFET

Metal-Oxide-Semiconductor

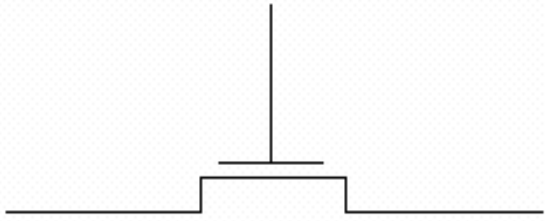
Field-Effect Transistor



MOSFET = switch

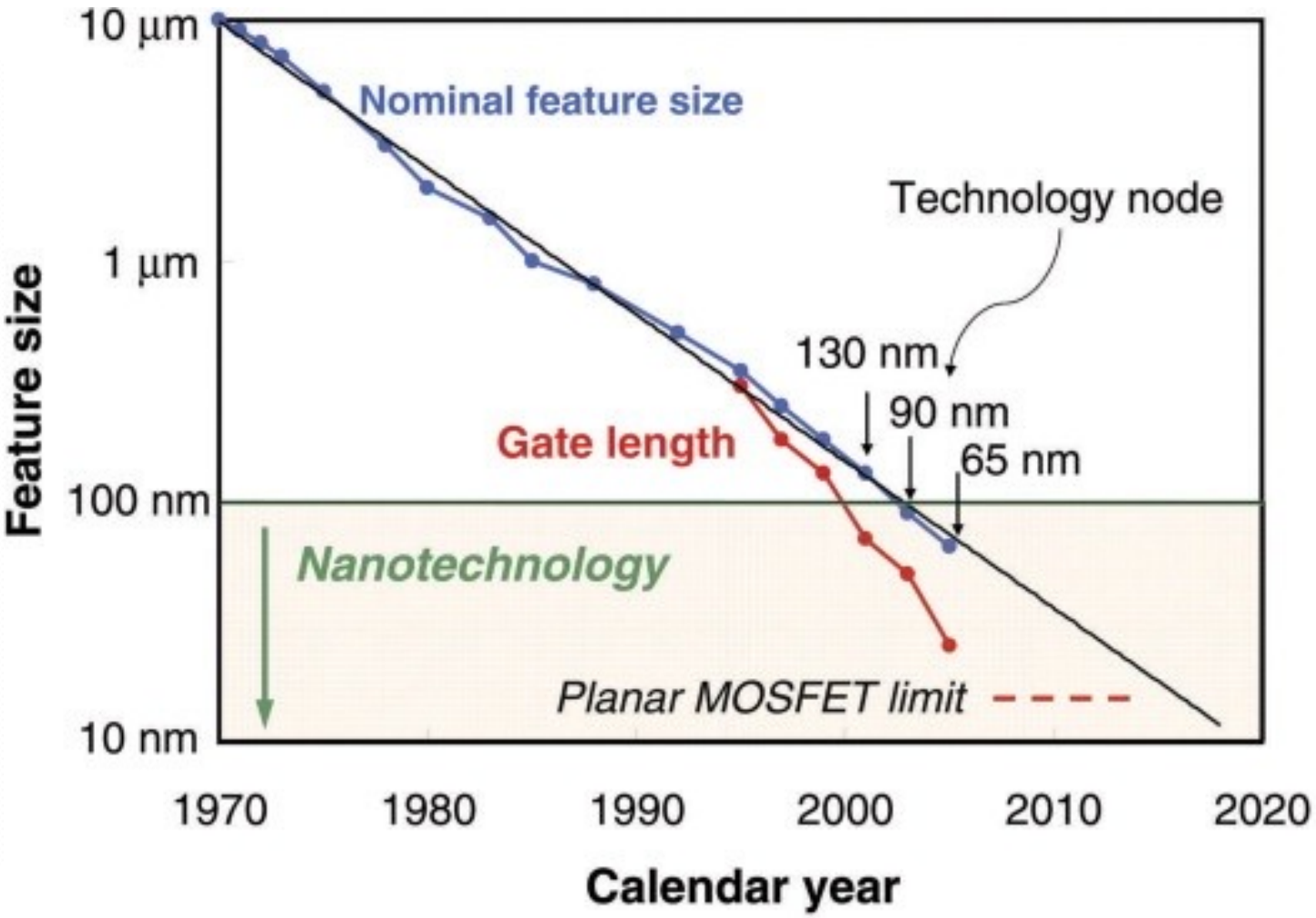


$V_G < V_T$ switch open

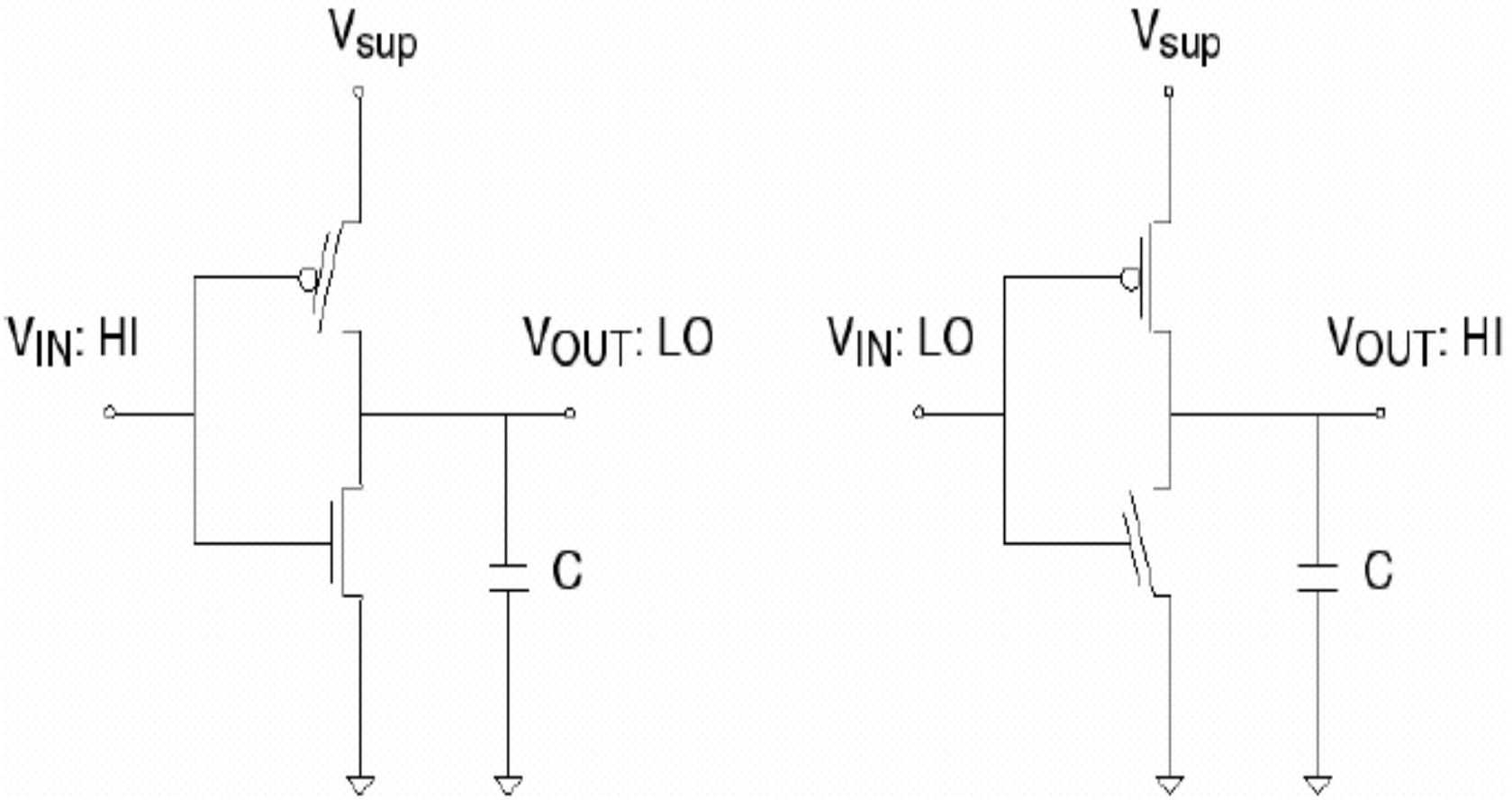


$V_G > V_T$ switch closed

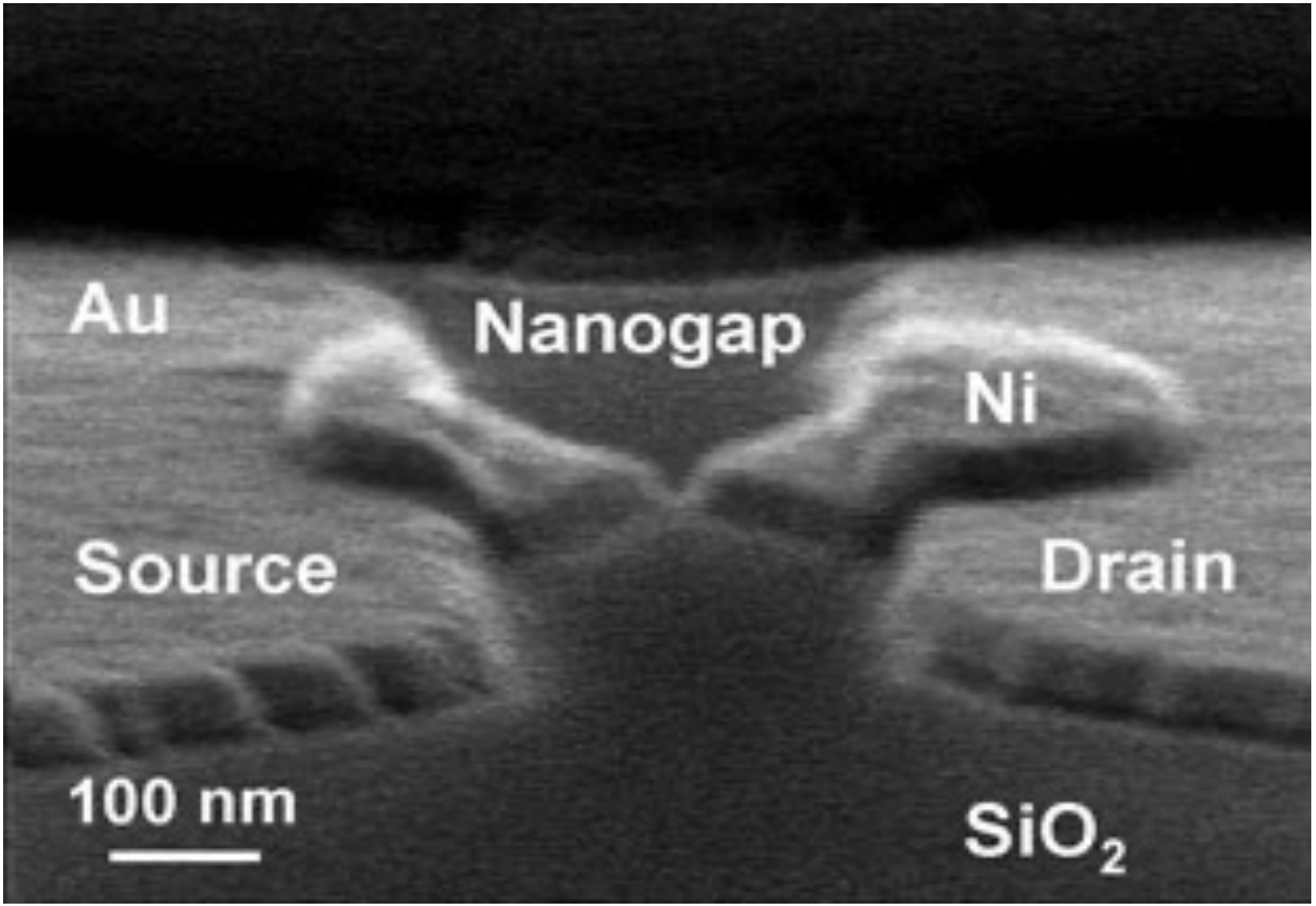
Keys to Success: Scaling



Keys to Success: The CMOS

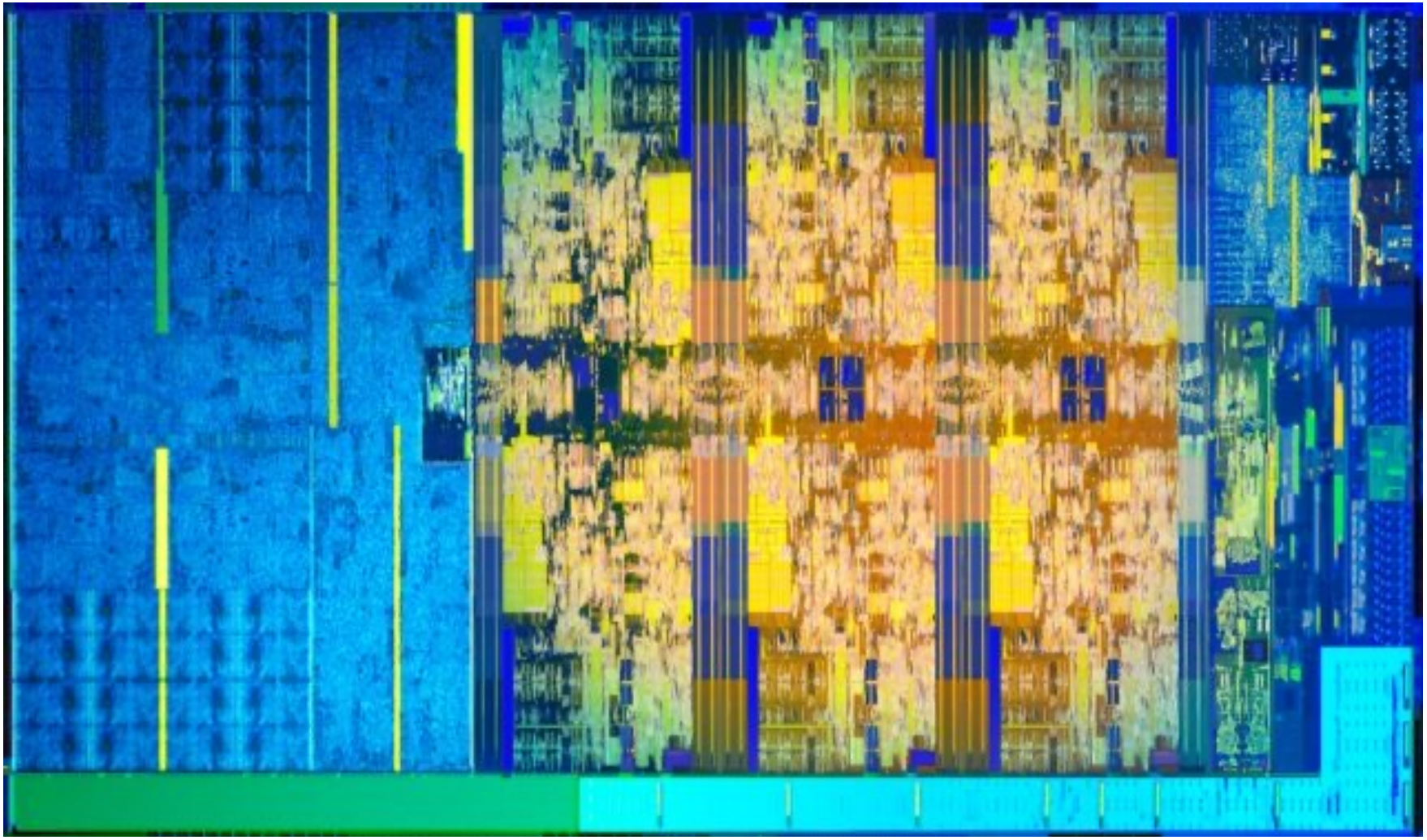


Keys to Success: Micro/Nano Fabrication

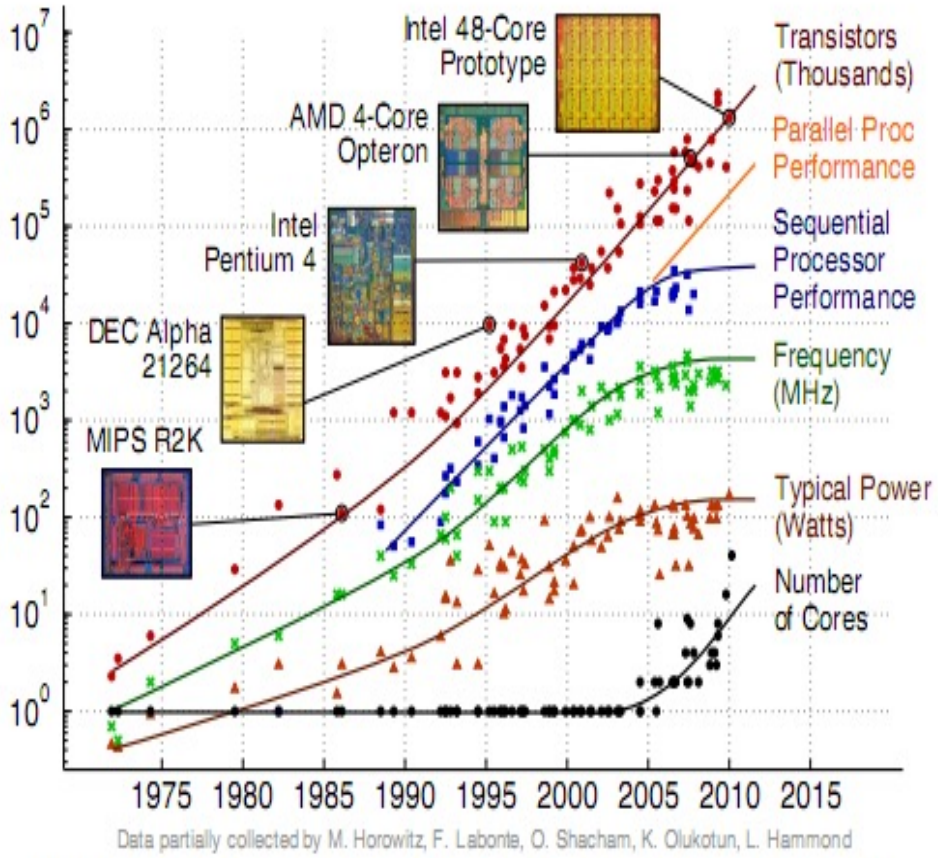


Keys to Success: Circuit Engineering

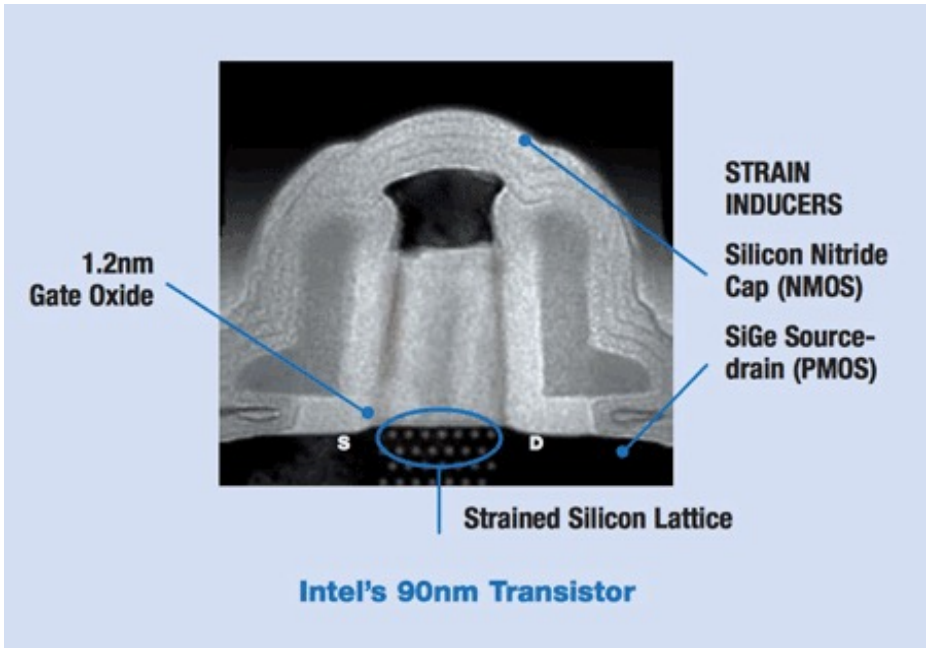
The Intel Core i9



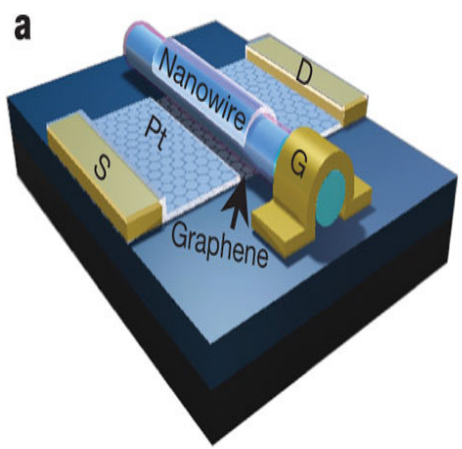
Technological Challenges: RC Delay and Power

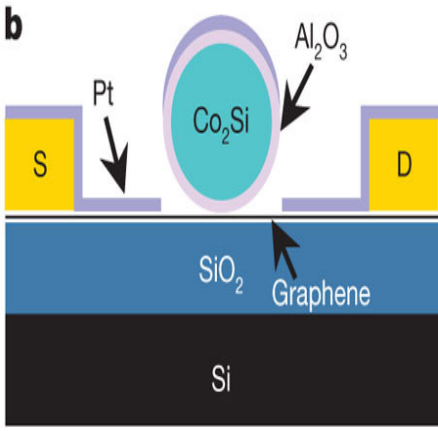


Prepared by C. Batten - School of Electrical and Computer Engineering - Cornell University - 2005 - retrieved Dec 12 2012 - http://www.csl.cornell.edu/courses/ece5950/handouts/ece5950-overview.pdf



Intel's 90nm Transistor





Syllabus

Nano Photonics and Plasmonics Research

Instructor Information	Mohammad Alsunaidi	Office M4-232	Office Hours MTWT: 13:00-14:00	Phone ---	Email: mohammad.alsunaidi@marmara	
Course Information	Text: Sedra/Smith, *Microelectronic Circuits 5th Ed.*	Grading	Class Work 15%	Lab Work 15%	Mid-term Exam 30%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Lab Activity	Exercises
1 Feb 12-16	---	**Introduction to Physics of Semiconductors.** Conduction in Semiconductors, N-type and P-type Semiconductors, The PN junction.	3	7	NL	(3) 112,113,114, 116,118.
2 Feb 19-23	---	**Diodes**. Introduction, The Ideal diode, Terminal characteristics of the Diode.	3	1-2	Experiment (1) LTSpice	(3) 2,3,4,6,9,10.
3 Feb 26-Mar 1	Course drop/add Week	**Diode Circuit Analysis**. Diode Models, Graphical and Analytical Diode Circuit Analysis.	3	3	Experiment (2) LTSpice	(3) 17,20,22,25,26, 40,41,46,47.
4 Mar 4-8	---	**Diode Applications**. The Zener diode. Half and Full-wave Rectifiers.	3	4-6	Experiment (3) Diode ckt Simulation	(3) 64,65,66,71,76, 78,82.
5 Mar 11-15	Quiz 1 March 12 (In-class)	**Diode Applications**. Limiting and Clamping Circuits, Voltage Doublers. Other Diode types. Examples	3	8	Experiment (4) Basic Diode Circuits	(3) 93,98,105.
6 Mar 18-22	---	**MOS Field-Effect Transistors (MOSFETs).** Device Structure and Modes of Operation. Types, Symbols and Conventions.	4	1-2	Experiment (5) Diode Rectifiers	(4) 4,6,12,15,19,23.
7 Mar 25-29	---	**MOSFET DC Circuits.** Biasing and Circuit DC Analysis.	4	3	Problem Session	(4) 35,36,37,41,42, 44,47.
April 1-7	**Midterm Exams period**					
		Apr 9-12 Holidays				
9 Apr 15-19	---	**MOSFET Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CS, CG and CD).	4	4-6	NL	(4) 63,69,74,75,77.
10 Apr 22-26	Midterm Make-up (Excused, Week 11)	**MOSFET Amplifiers.** Single-stage Amplifiers (CS, CG and CD).	4	7	Experiment (6) MOSFET DC Ckts	(4) 81,85,88.
11 Apr 29-May 3	Labor Day: May 1	**Bipolar Junction Transistors (BJTs).** Structure and Modes of Operation, Types, Symbols and Conventions,	5	1-2	NL	(5) 3,4,7,13,20, 31,39.
12 May 6-10	---	**BJT DC Circuits**. Biasing. Current-voltage (IV) Characteristics. BJT as a switch.	5	4	Experiment (7) MOSFET Amplifiers	(5) 69,71,72,74, 79,82,84,86.
13 May 13-17	Quiz 2 May 14 (In-class)	**BJT Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CE, CB and CC).	5	6	Experiment (8) BJT Circuits	(5) 104,106,107, 115,116,124.
14 May 20-24	YS Day May 19	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	NL	(5) 130,131,134, 137,141,143.
15 May 27-31	---	**CMOS Logic Circuits.** Overview, the CMOS Inverter, Other CMOS Logic circuits, CMOS transistor sizing	10	1-3	Experiment (9) BJT Amplifiers	(10) 18,25,28.
June 3-14	**Final Exams period**	**July 1-7 (Re-sit Final Exams period)**				

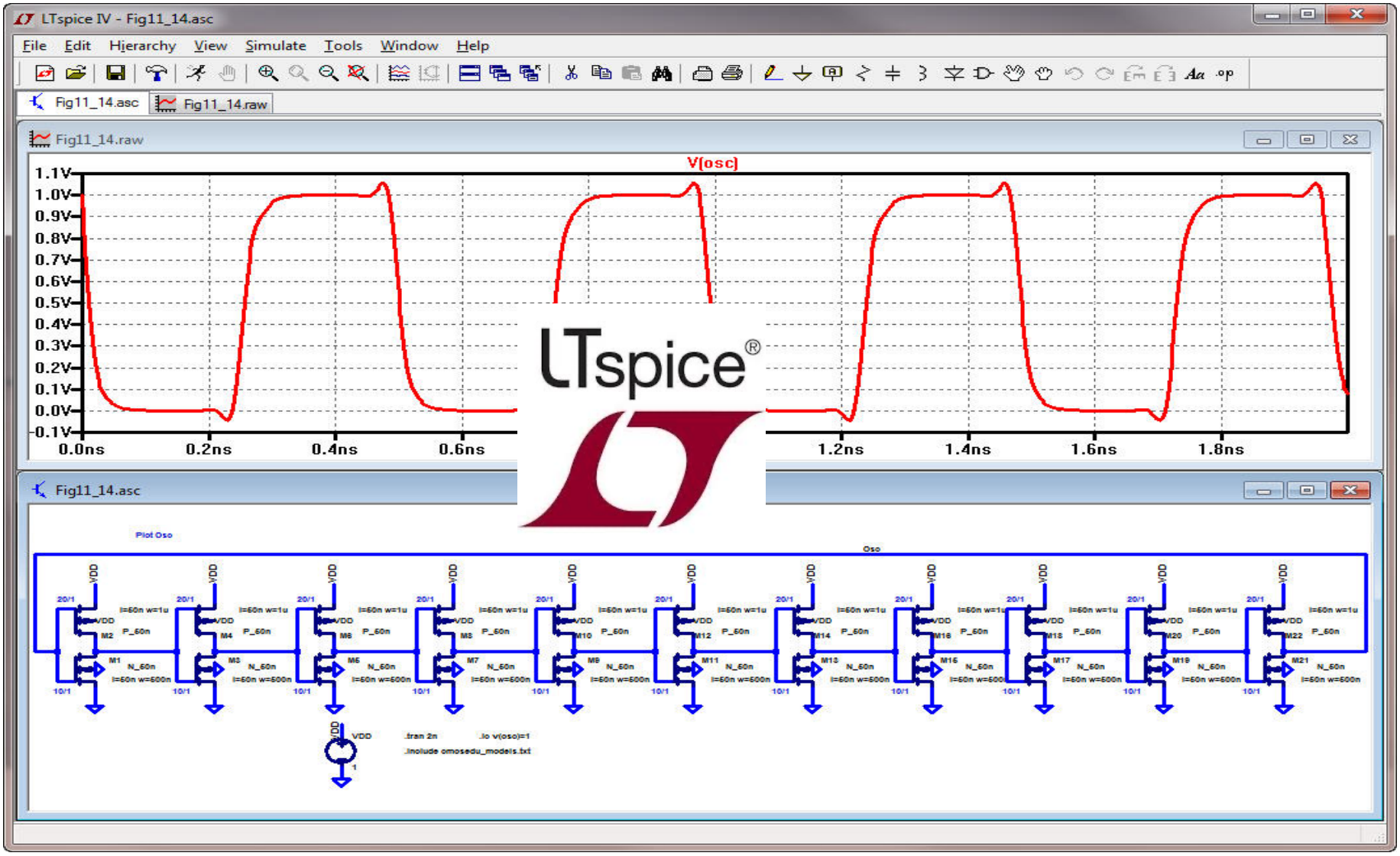
Pre-requisite Material

Physics II: Charges, Electric field.

Math I: Simple integration and differentiation.

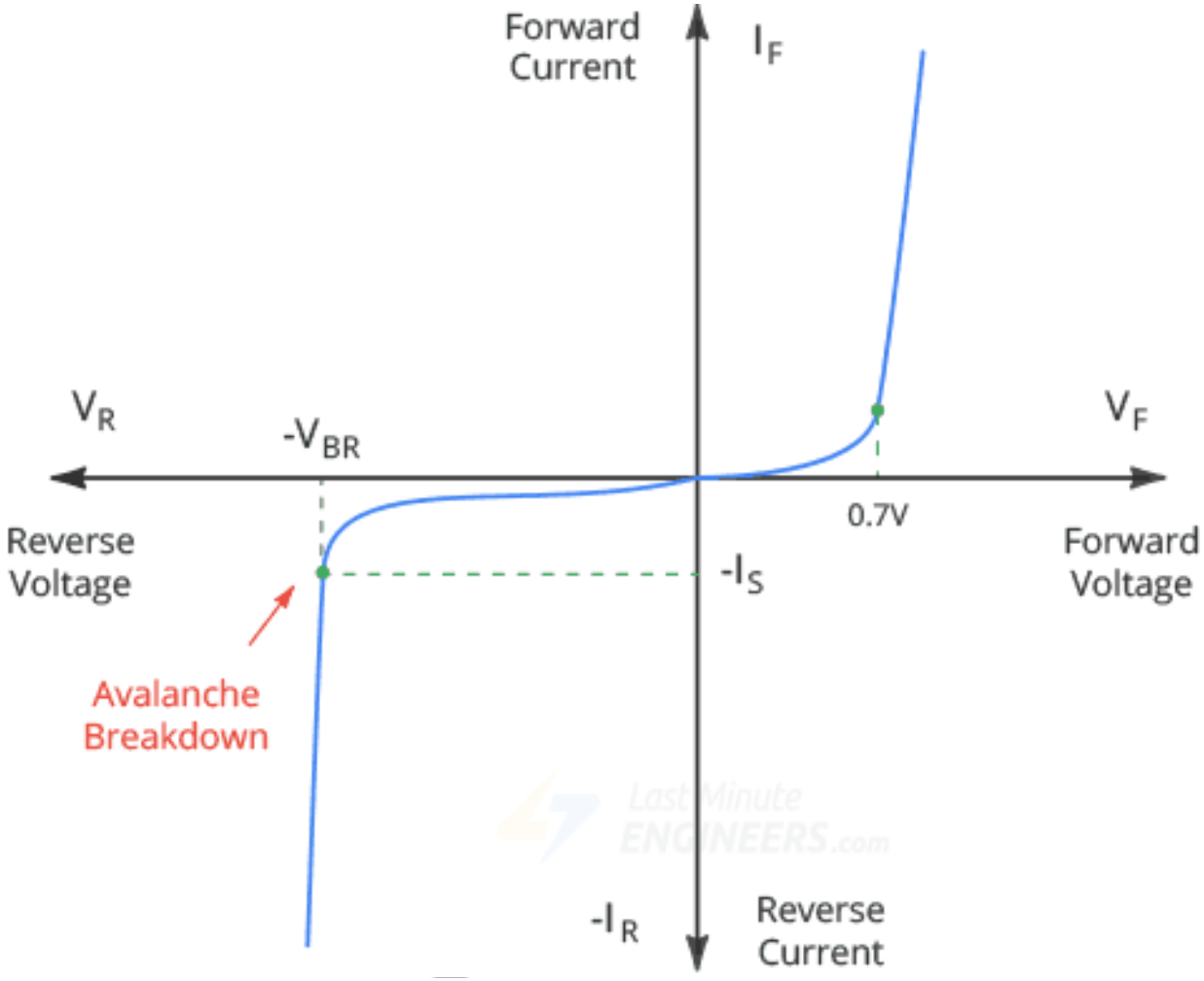
Circuits I: A lot of circuit analysis, including
the concepts of voltage, current and power.
Ohm's law
KVL and KCL equations.
Voltage and current dividers,
Thevenin's theorem,
Dependent sources, etc.

Simulation Software: LTspice

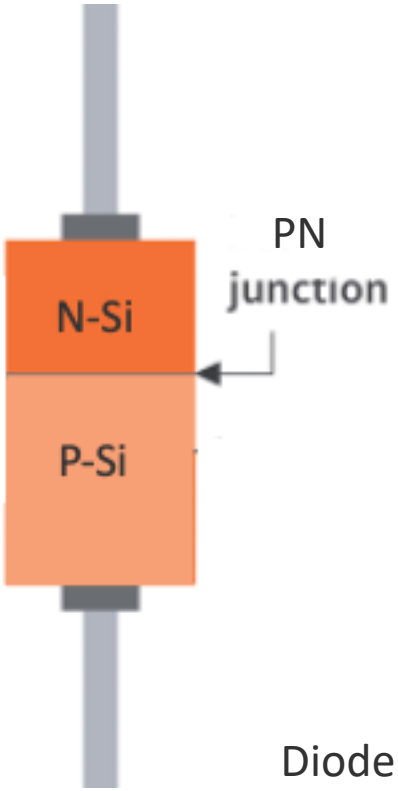


End of Part I

Transistor Structure



Half a Transistor
One PN junctions



Diode

Semiconductors Fundamentals

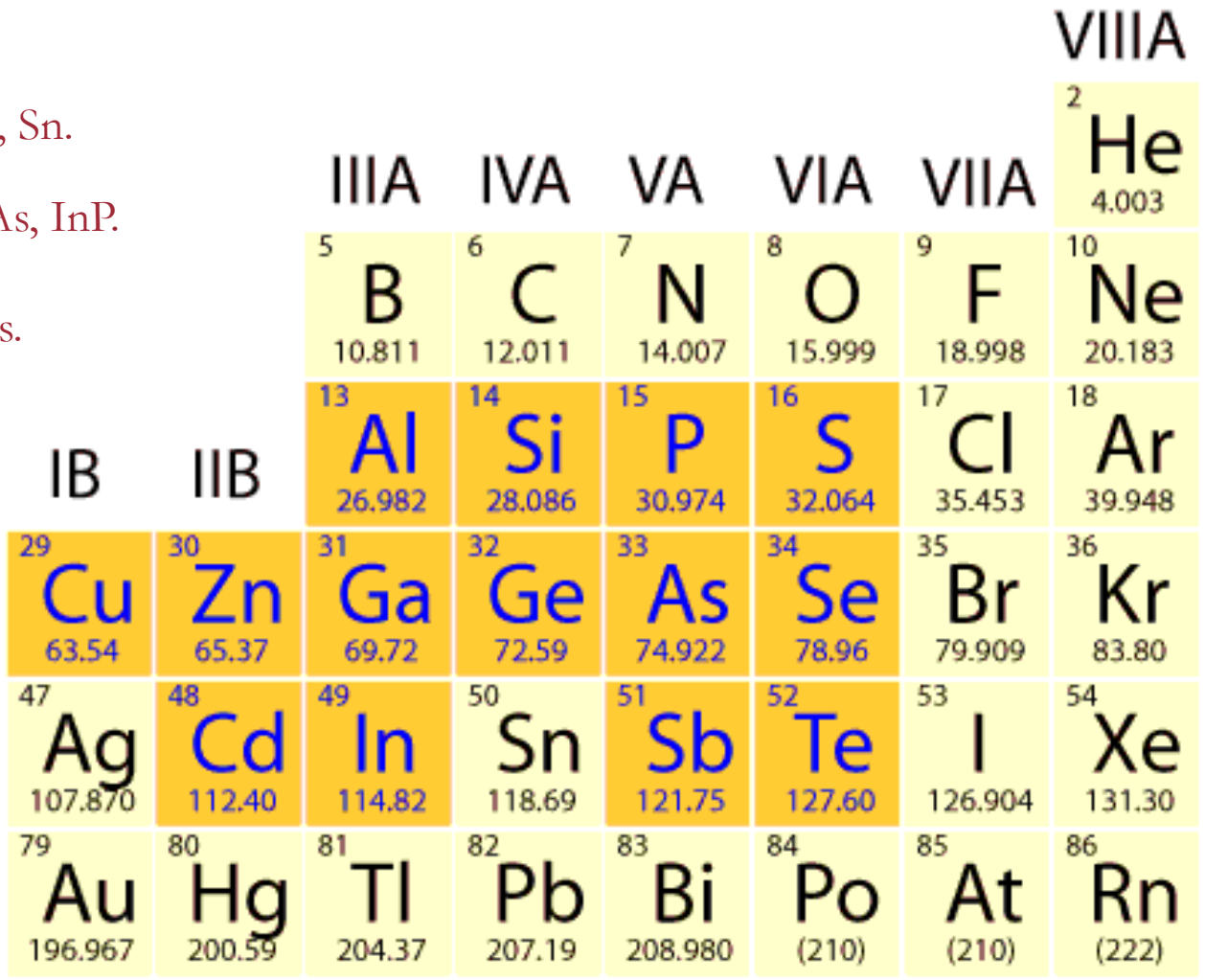


Semiconductors

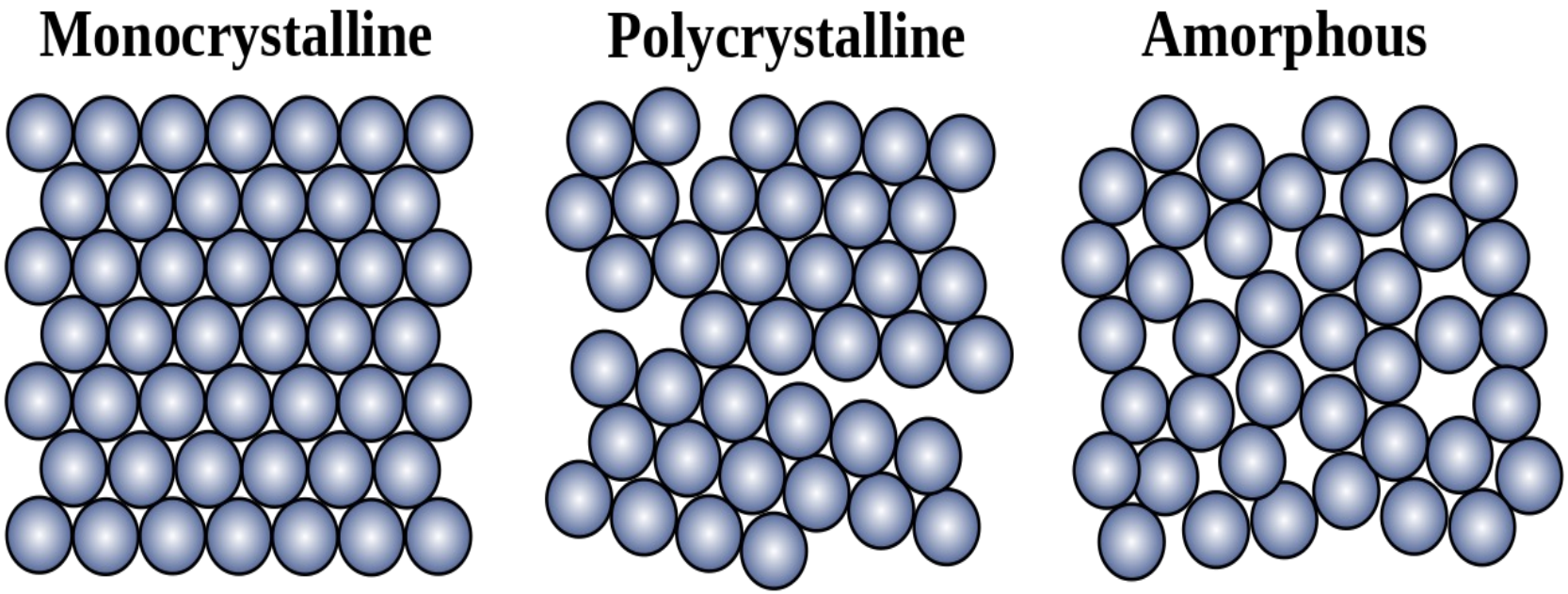
Elemental: Si, Ge, Sn.

Compounds: GaAs, InP.

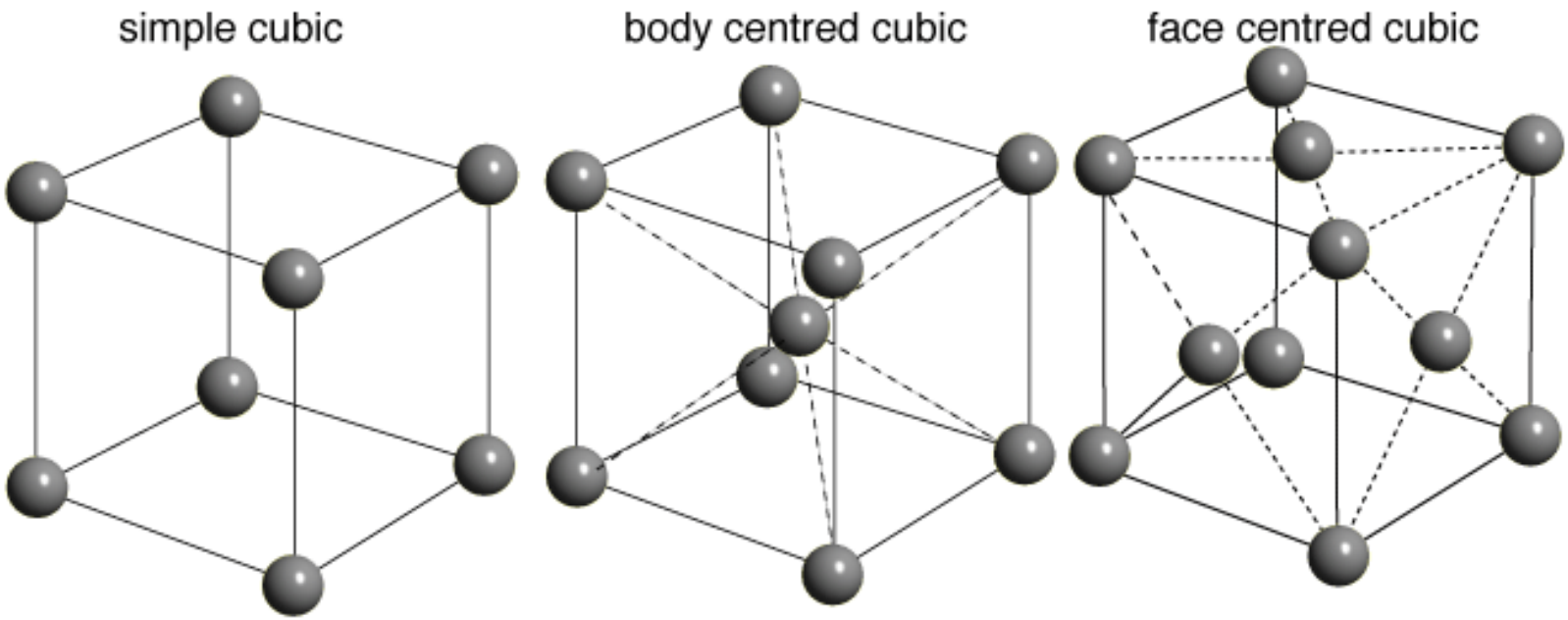
Alloys: $Al_{1-x}Ga_xAs$.



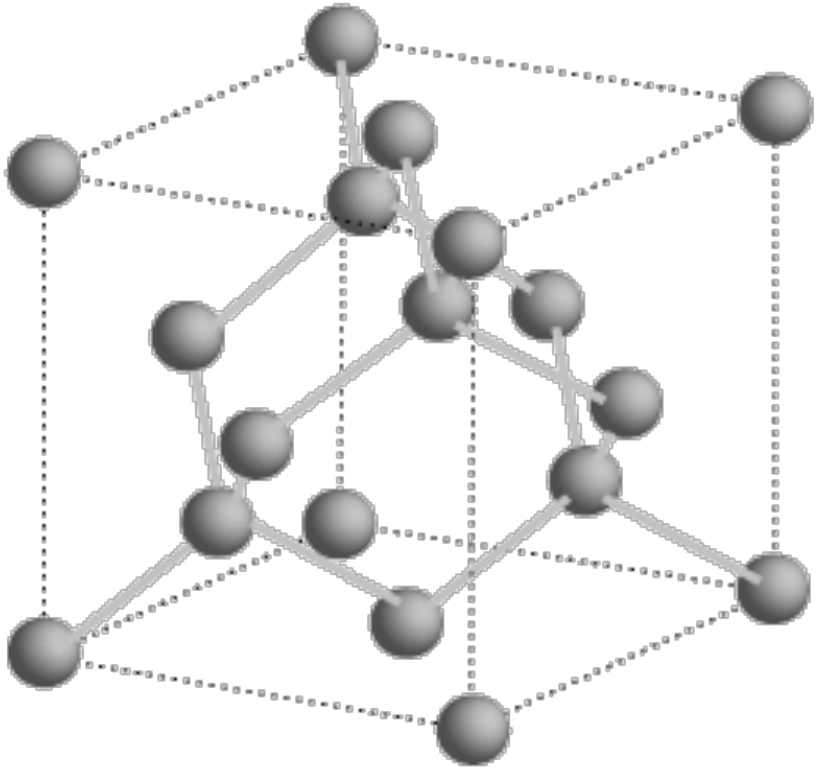
Structure



Crystal Structures: Cubic

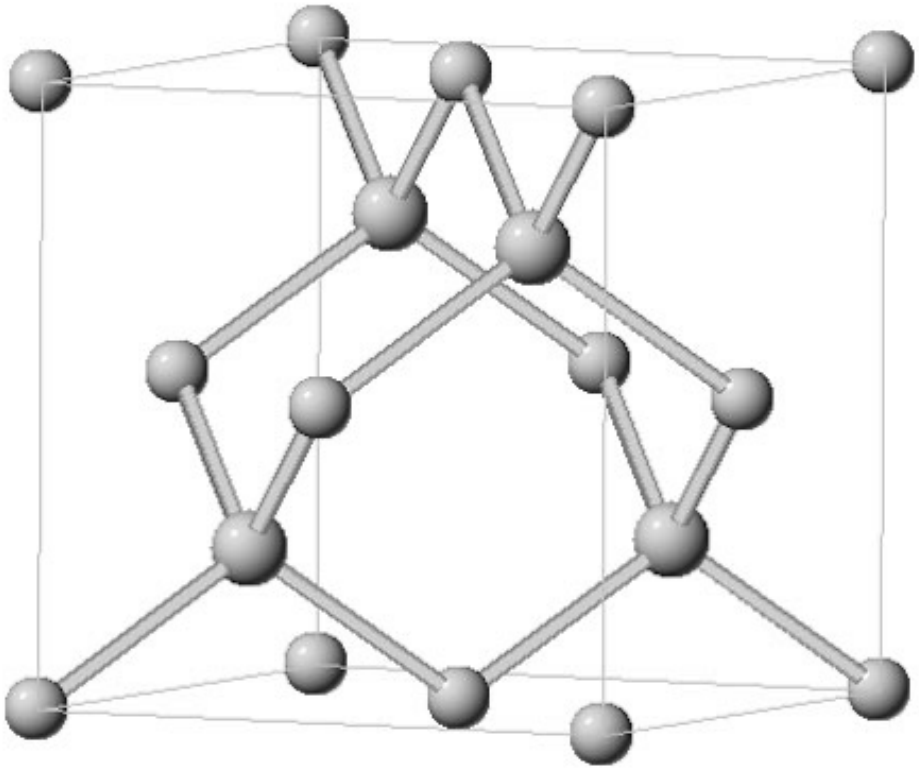


Crystal Structures: Diamond

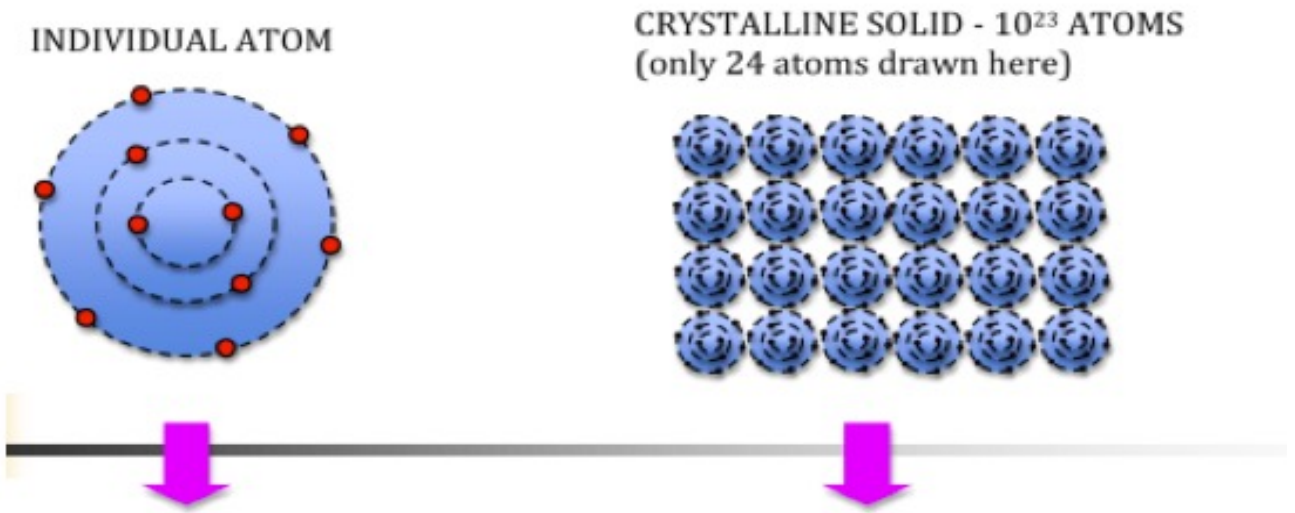


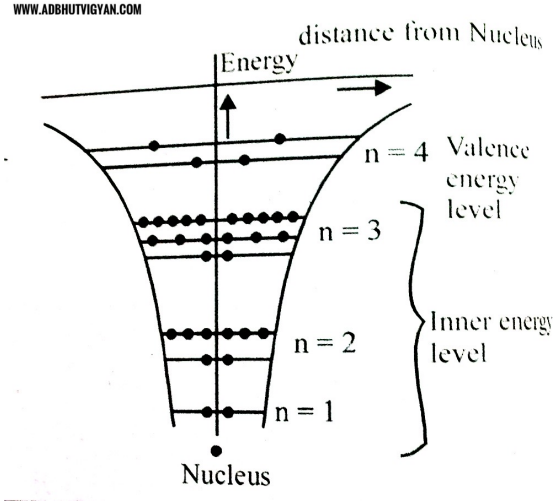


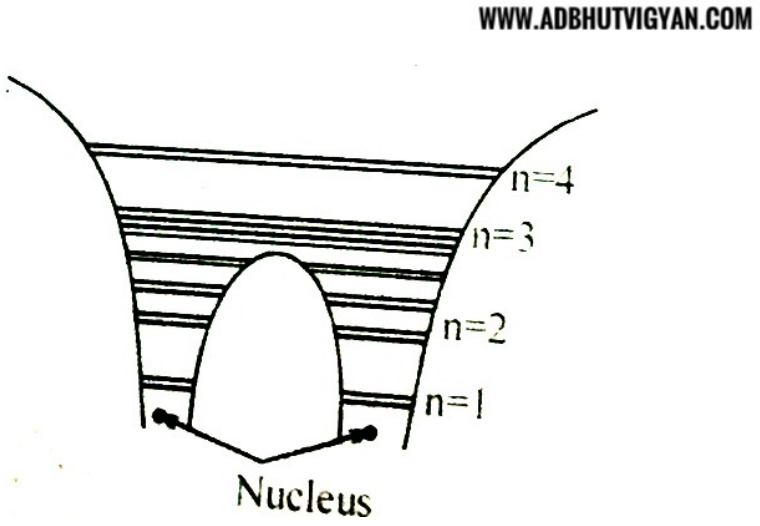
Crystal Structures: Zinc-blende



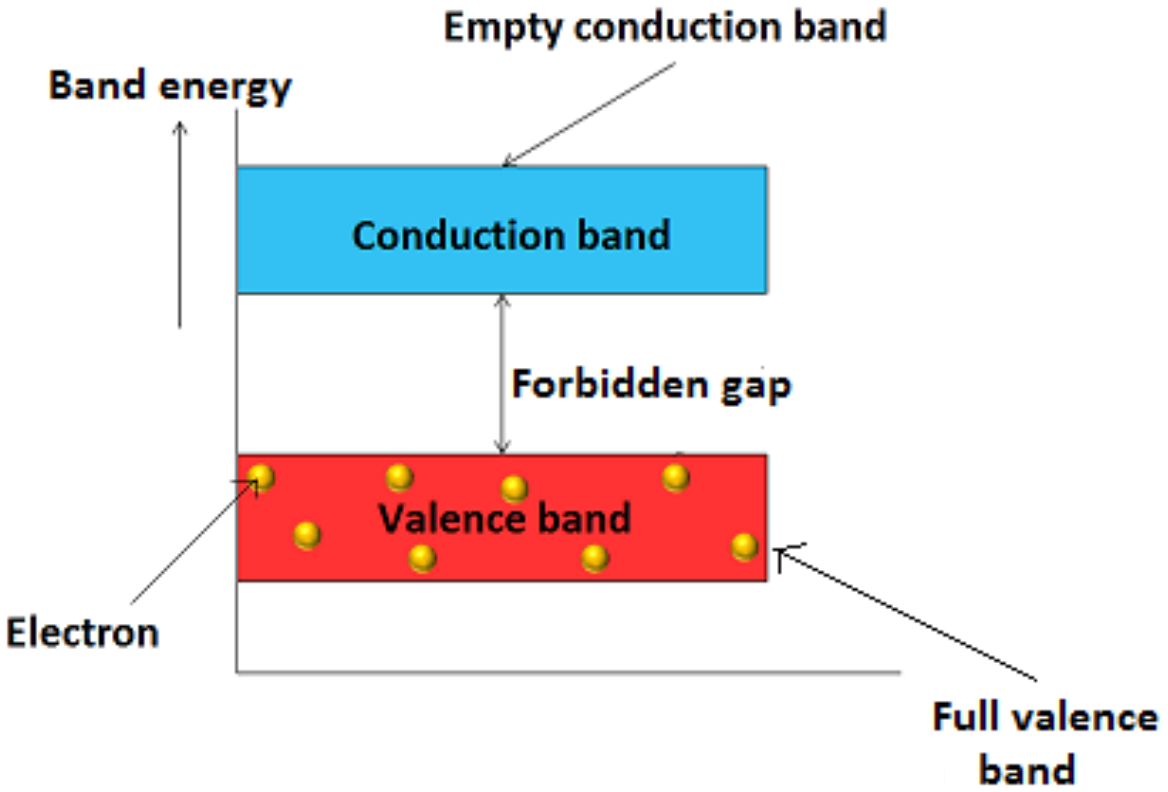
Energy Bands: Isolated atoms and Crystals



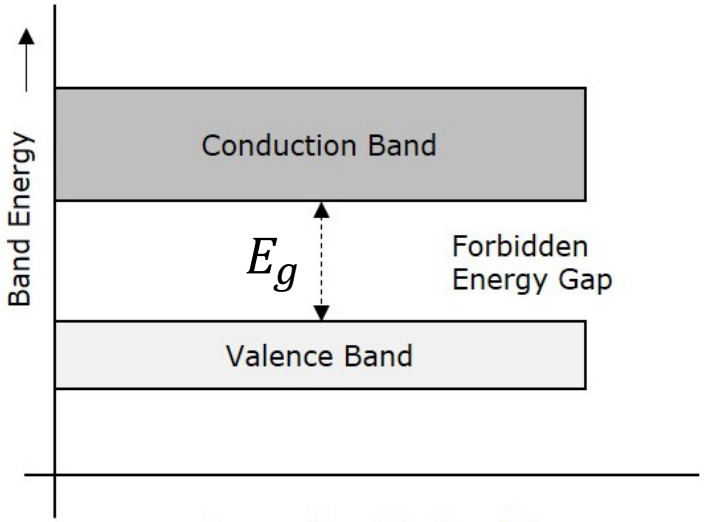




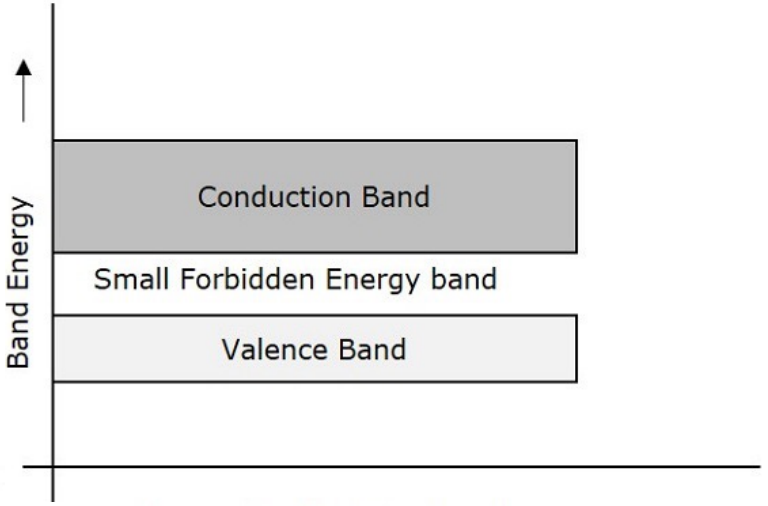
Energy Bands: Crystals



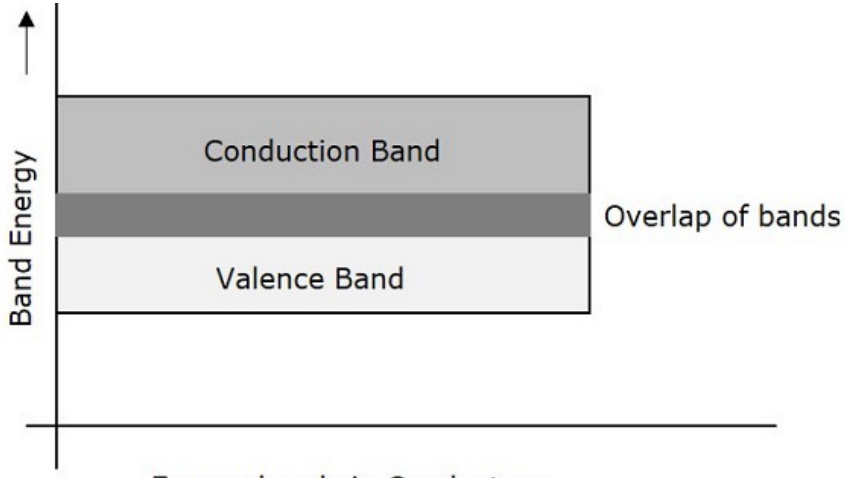
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

Diamond $E_g = 6$ eV

Silicon $E_g = 1.12$ eV

Copper $E_g = 0$ eV

Common Semiconductors

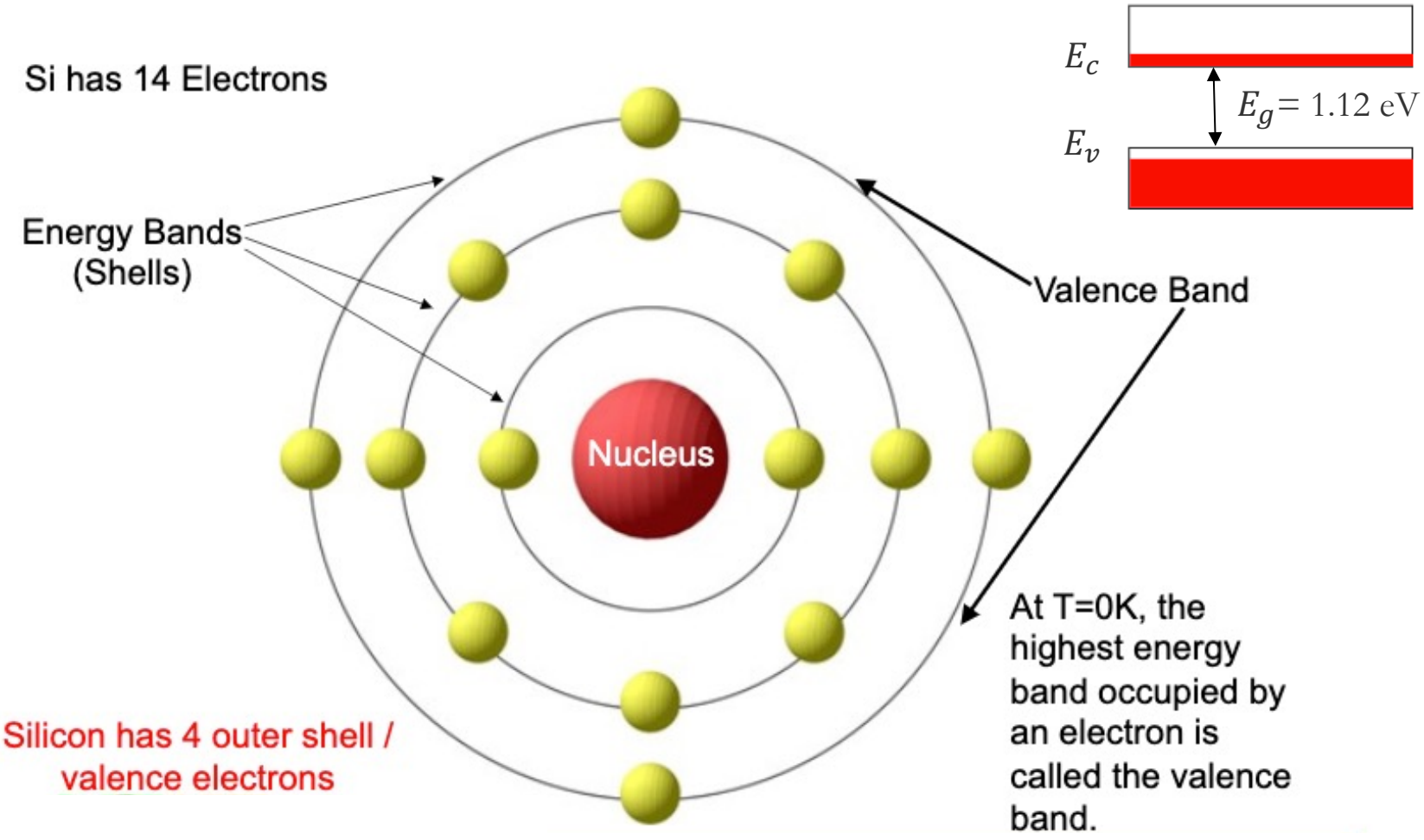
Semiconductor	Bandgap Energy E_G (eV)
Carbon (diamond)	5.47
Silicon	1.12
Germanium	0.66
Tin	0.082
Gallium arsenide	1.42
Gallium nitride	3.49
Indium phosphide	1.35
Boron nitride	7.50
Silicon carbide	3.26
Cadmium selenide	1.70

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

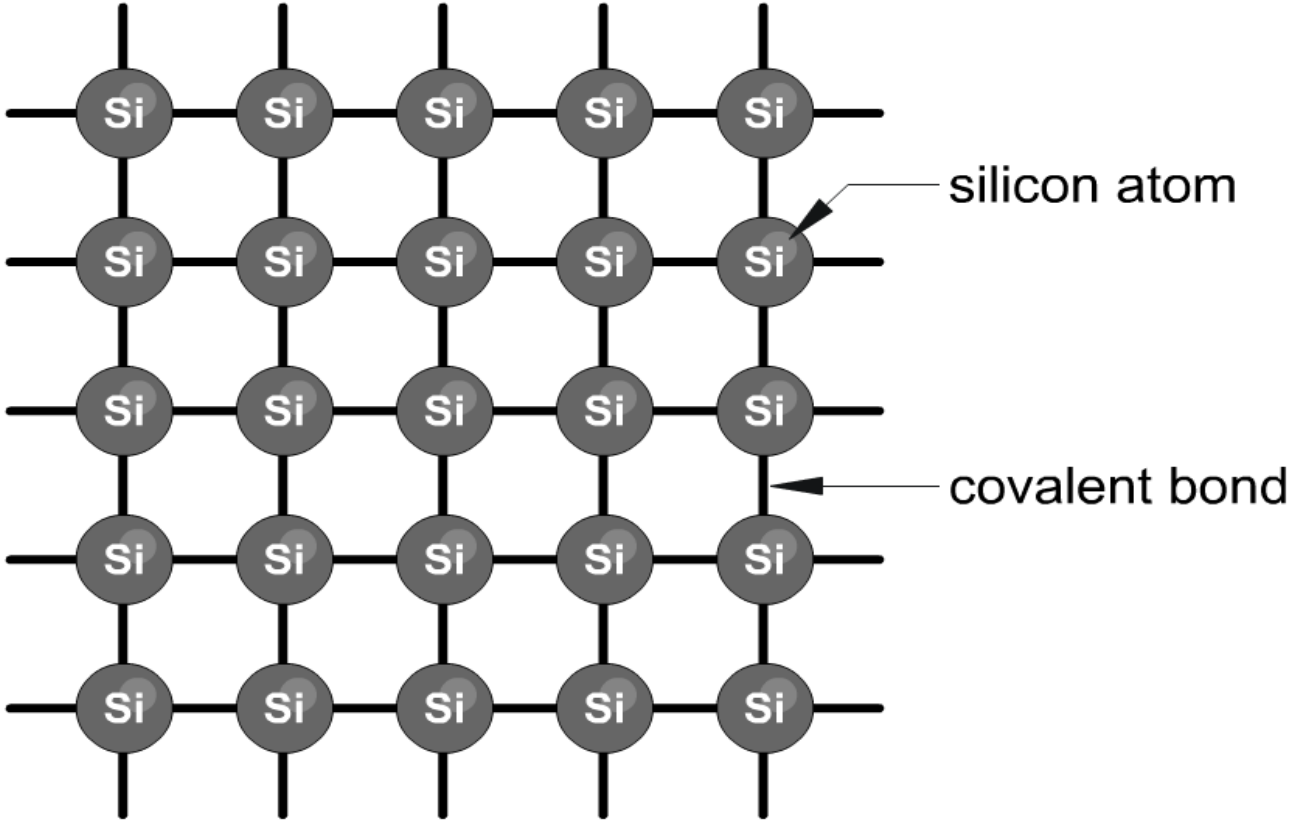
Example: Silicon



The Silicon Atom



Covalent Bonding in Pure Si



How many *free* electrons?

Carrier Count: Intrinsic Si

The density of free electrons is given by

$$n_i^2 = N_c N_v \, e^{-(E_c - E_v)/kT} = N_c N_v e^{-E_g/kT}$$

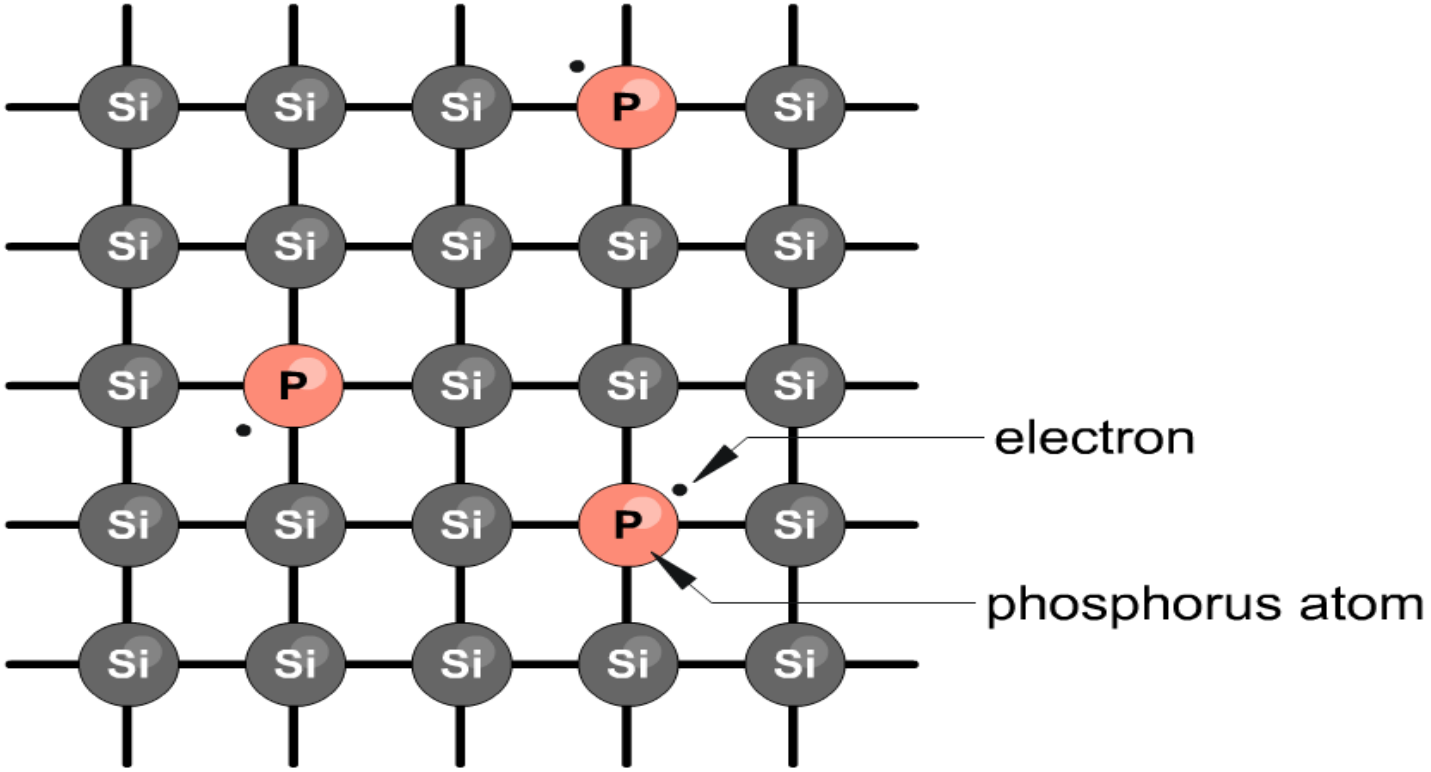
- N_c: Density of states in conduction band
- N_v: Density of states in valence band
- E_g: Band gap energy in eV
- k: Boltzmann constant 8.62×10^{-5} eV/K
- T: Temperature in K

For Silicon $E_g = 1.12$ eV $N_c = 3.2\times10^{19}$ cm^{-3} $N_v = 1.8\times10^{19}$ cm^{-3}

At 300 K $n_i = 1\times10^{10}$ cm^{-3}

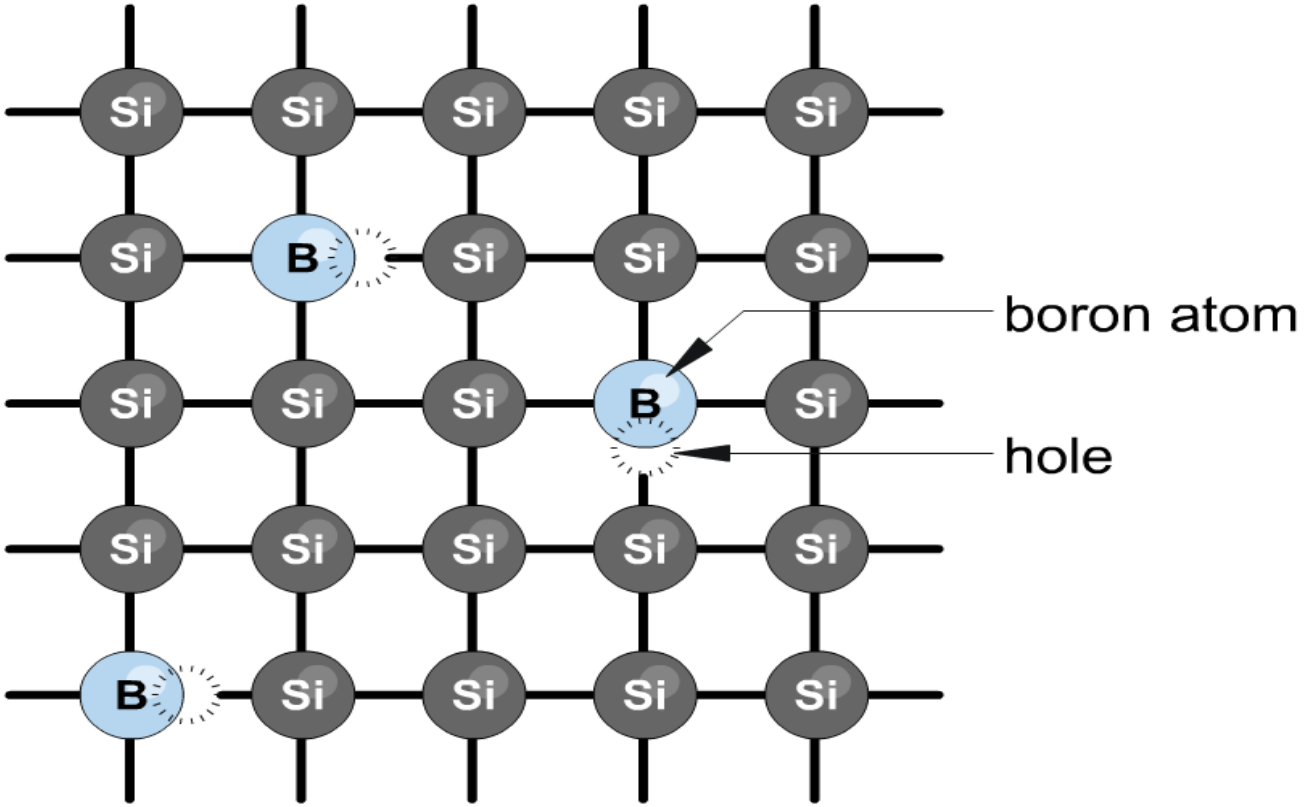
For a good conductor At 300 K $n_i = 1\times10^{23}$ cm^{-3}

Doping: N-type Si



Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si



Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Carrier Count: Doped Si

The density of free electrons is N-type silicon is given by

$$n \approx N_D$$

The density of holes electrons is P-type silicon is given by

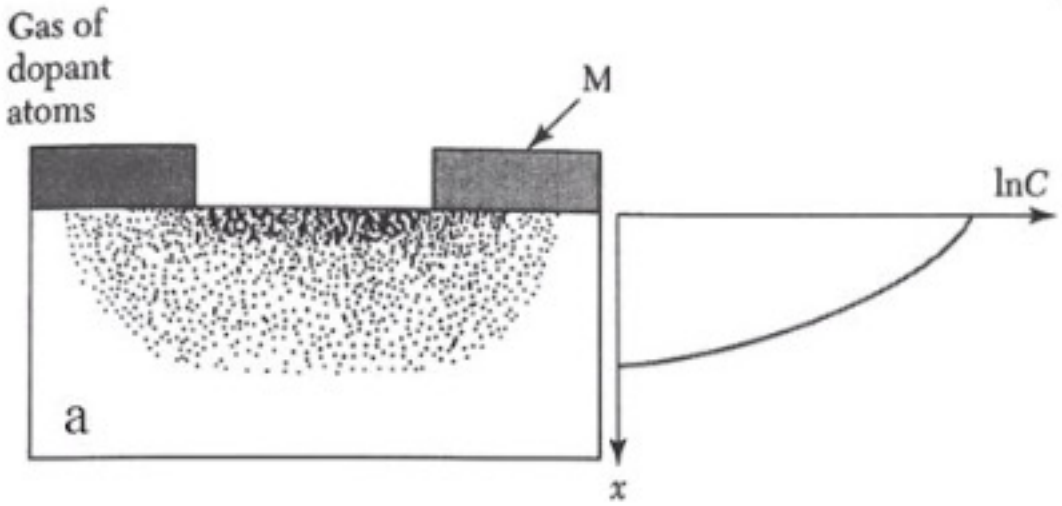
$$p \approx N_A$$

Does this doping change the original material?

Compare the number of original atoms to the number of acceptor/donor atoms.

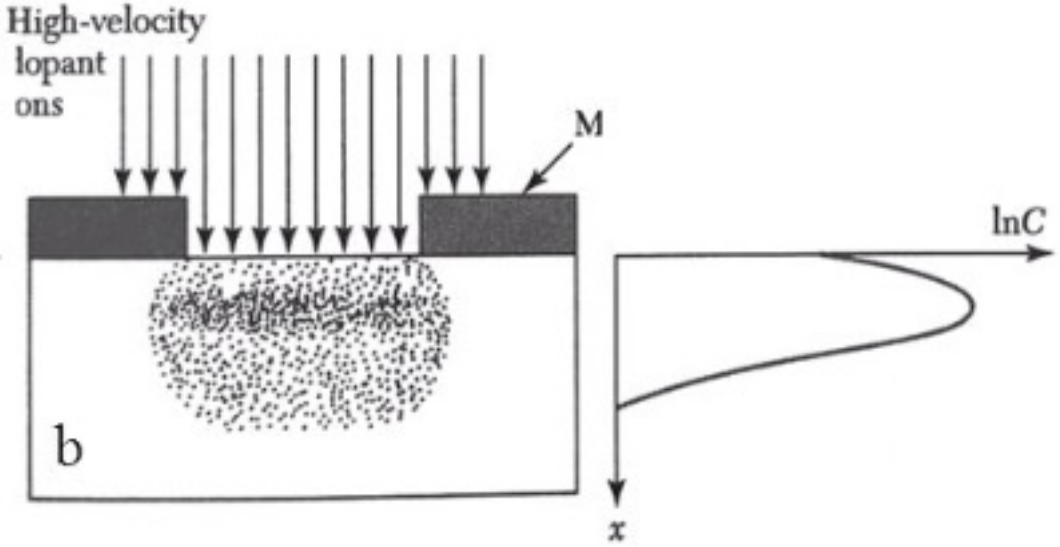
On average, the density of original atoms is about $10^{23} - 10^{24}$ atoms/cm^3.

Methods of Doping: Diffusion



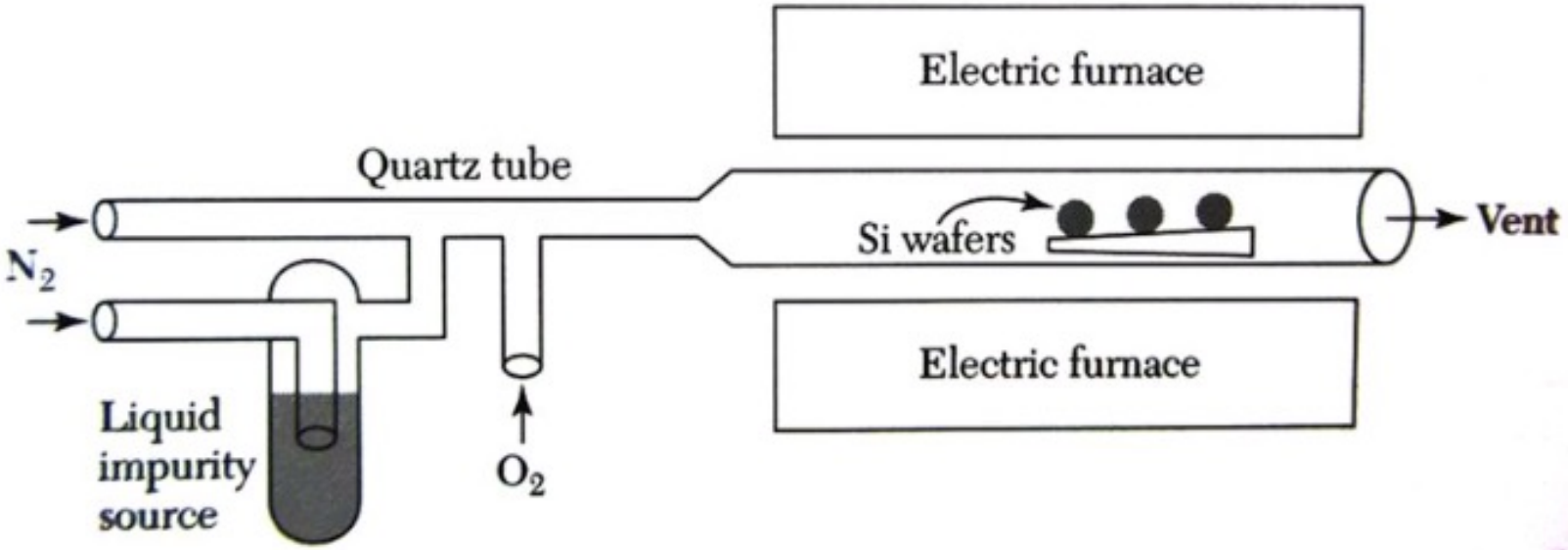
Diffusion is a cheaper and more simplistic method, but can only be performed from the surface of the wafers. Dopants also diffuse unevenly, and interact with each other altering the diffusion rate.

Methods of Doping: Ion implantation



Ion implantation is more expensive and complex. It does not require high temperatures and also allows for greater control of dopant concentration and profile. It is an anisotropic process and therefore does not spread the dopant implant as much as diffusion. This aids in the manufacture of self-aligned structures which greatly improve the performance of MOS transistors.

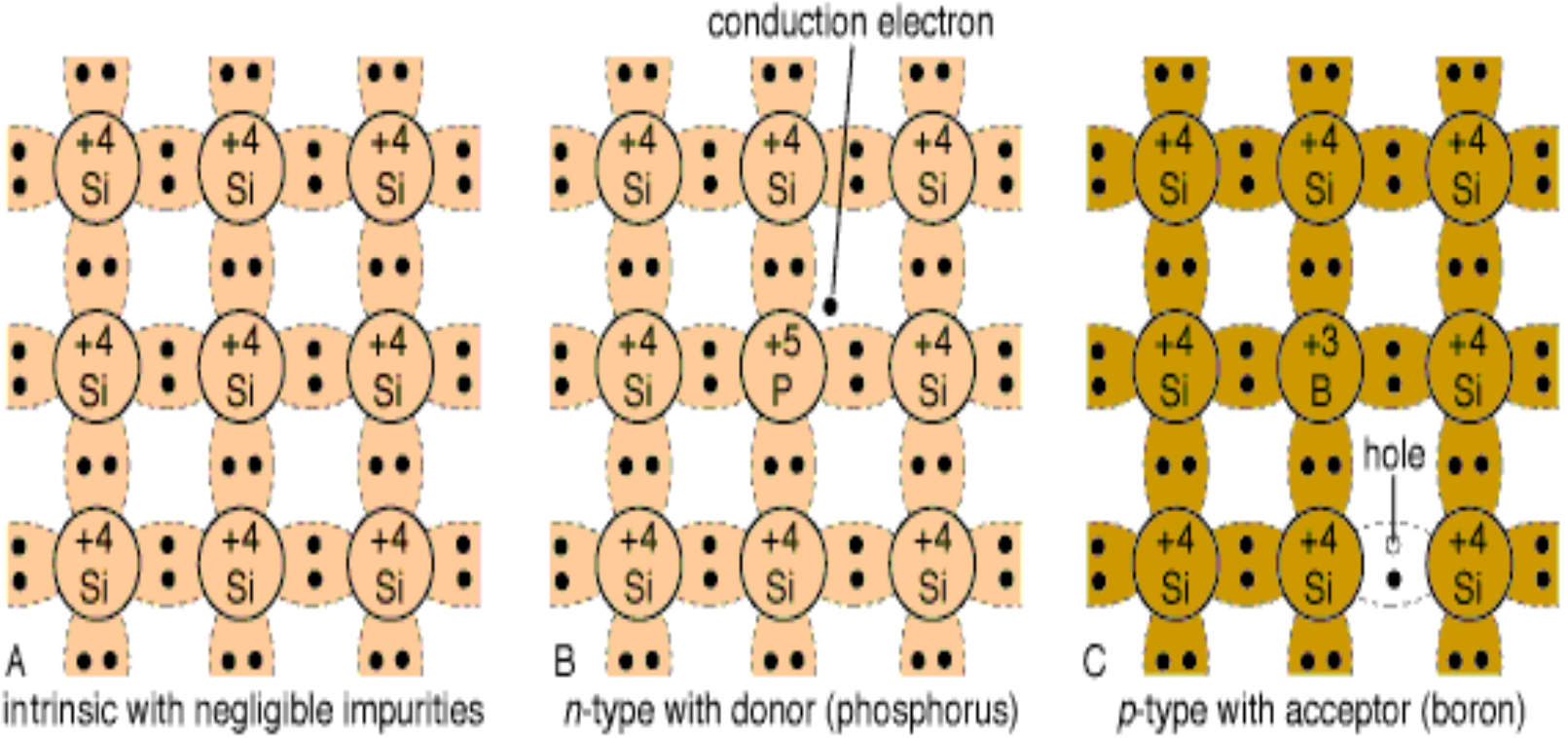
Example: N-type Si (Phosphorus doping)



$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$

$$2P_2O_5 + 5Si \rightarrow 4P + 5SiO_2$$

Silicon Doping



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The PN Junction

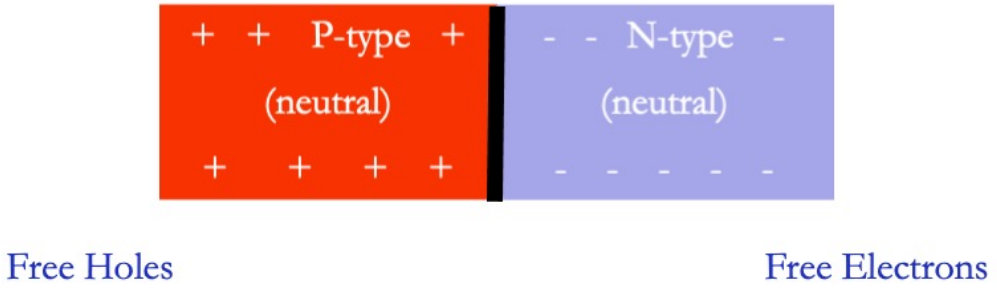
The PN Junction

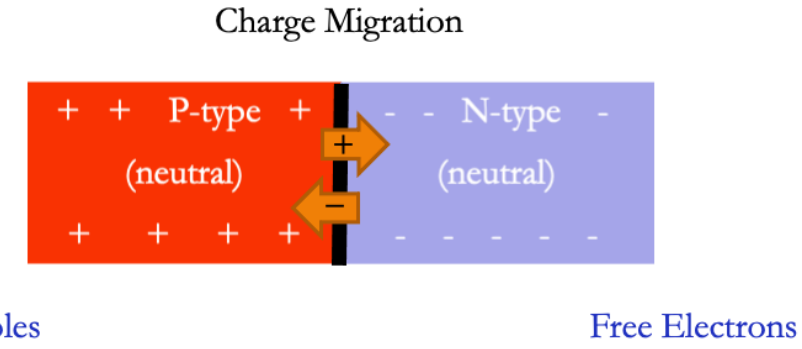


Formation of the PN Junction

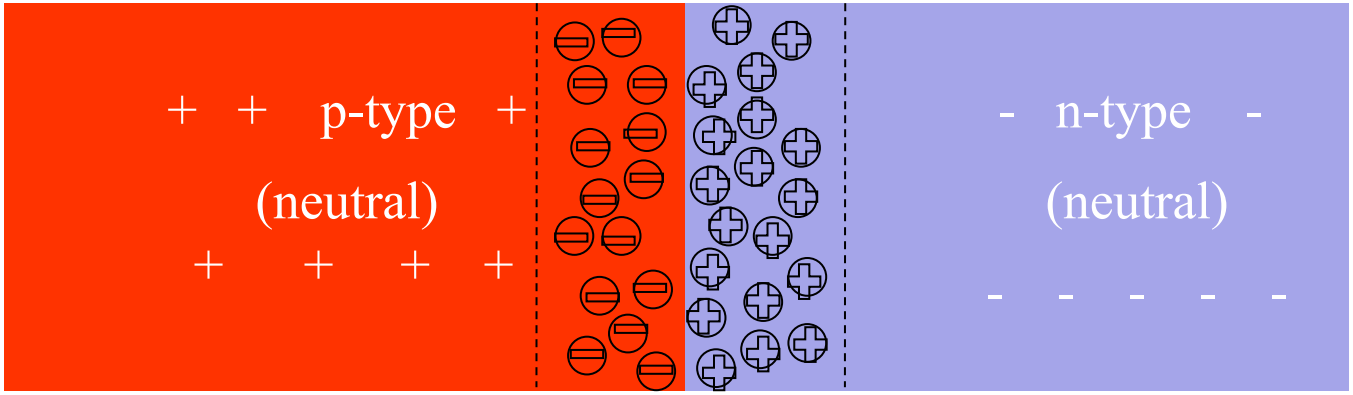
The basic structure of almost all active devices.

Diodes, transistors, lasers, photocells, solar cells, etc.

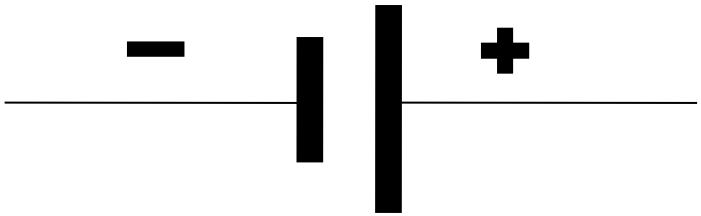




The PN junction: The Diode

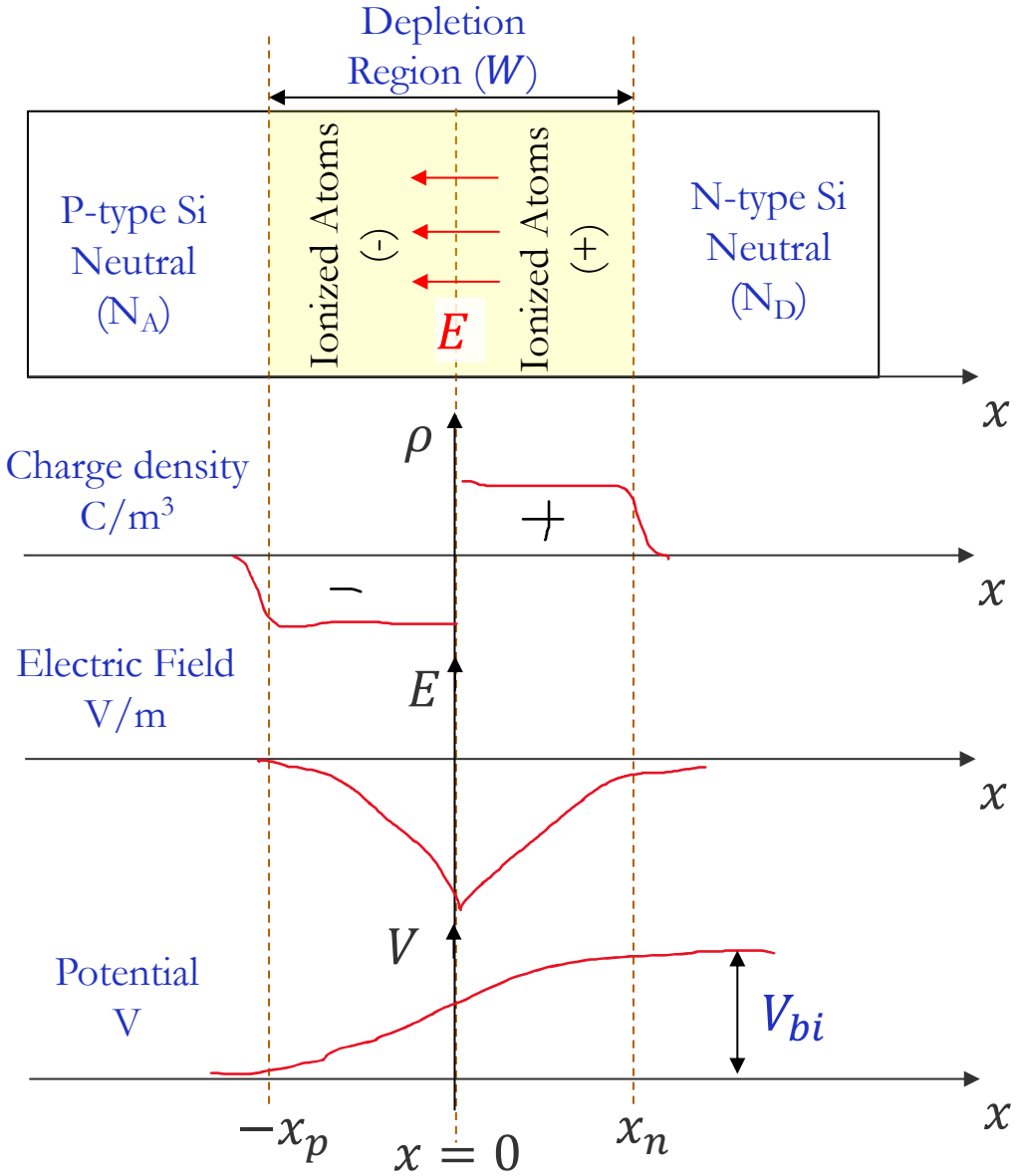


Built-in Voltage (V_{DO})





The PN Junction Parameters



$$V_{bi} = \frac{kT}{q}\ln\frac{N_D N_A}{n_i^2} \qquad W = x_n + x_p$$

$$x_n = \sqrt{\frac{2\varepsilon V_{bi}}{q}\frac{N_A}{N_D(N_A+N_D)}}$$

$$x_p = \sqrt{\frac{2\varepsilon V_{bi}}{q}\frac{N_D}{N_A(N_A+N_D)}}$$

$$E(x) = \frac{-qN_A}{\varepsilon}(x_p + x) \quad \text{p-side}$$

$$E(x) = \frac{-qN_D}{\varepsilon}(x_n - x) \quad \text{n-side}$$

$$V(x) = \frac{qN_A}{2\varepsilon}(x_p + x)^2 \quad \text{p-side}$$

$$V(x) = \frac{-qN_D}{2\varepsilon}(x_n - x)^2 + V_{bi} \quad \text{n-side}$$

$$C_j = \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon(V_{bi}-V_A)}{q}\frac{N_A+N_D}{N_A N_D}}}$$

The PN Junction Parameters

Example (1): Consider a Si pn junction at room temperature with donor doping of $N_D = 10^{16}$ cm^{-3} and accepter doping of $N_A = 10^{15}$ cm^{-3}. Calculate junction width.

At 300 K $\quad n_i = 1\times10^{10}$ cm^{-3} $\quad V_{bi} = 0.633$ V

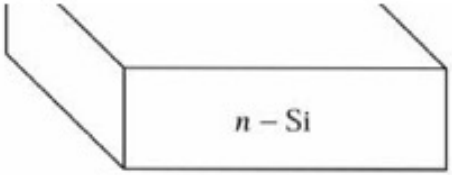
$x_p = 0.885\times10^{-6}$ m $\quad x_n = 0.0885\times10^{-6}$ m $\quad W = 0.973\ \mu$m

Example (2): For the junction in example 1, find the built-in voltage for $N_D = 10^{15}$ cm^{-3} and $N_D = 10^{17}$ cm^{-3}.

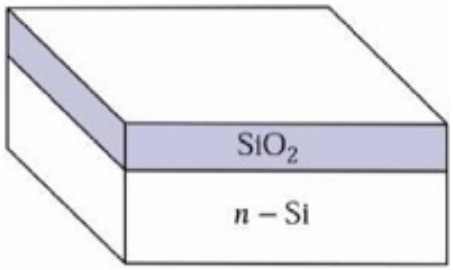
$$V_{bi} = 0.025 \ln \frac{N_D 10^{15}}{(10^{10})^2}$$

N_D (cm^{-3})	V_{bi} (V)
10^{15}	0.599
10^{16}	0.633
10^{17}	0.718

The PN junction: Fabrication

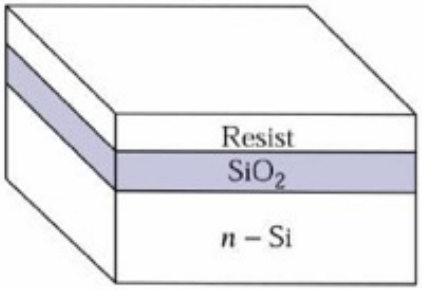


(a) Si wafer

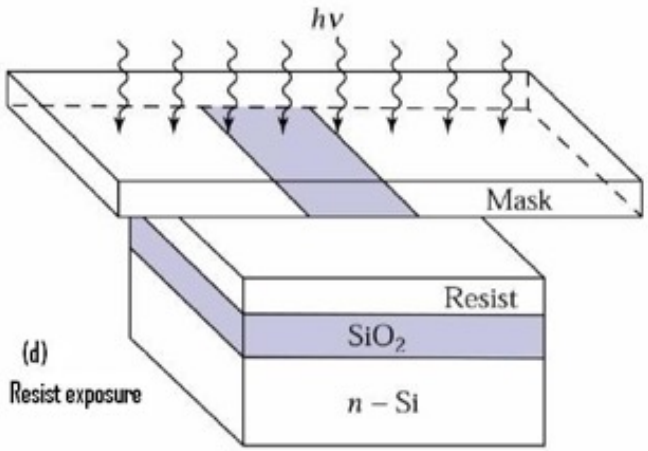


(b) Oxidation

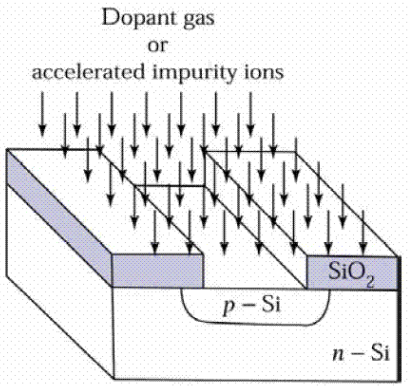




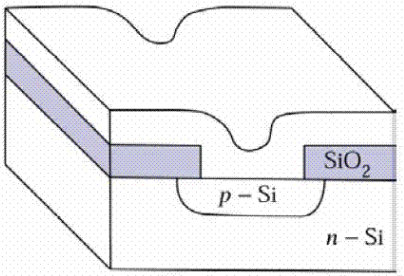
(c) Application of resist



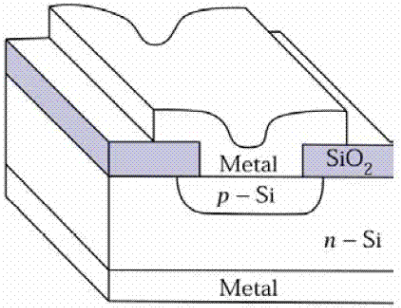
(d)
Resist exposure



(h) Diffusion or Implantation process

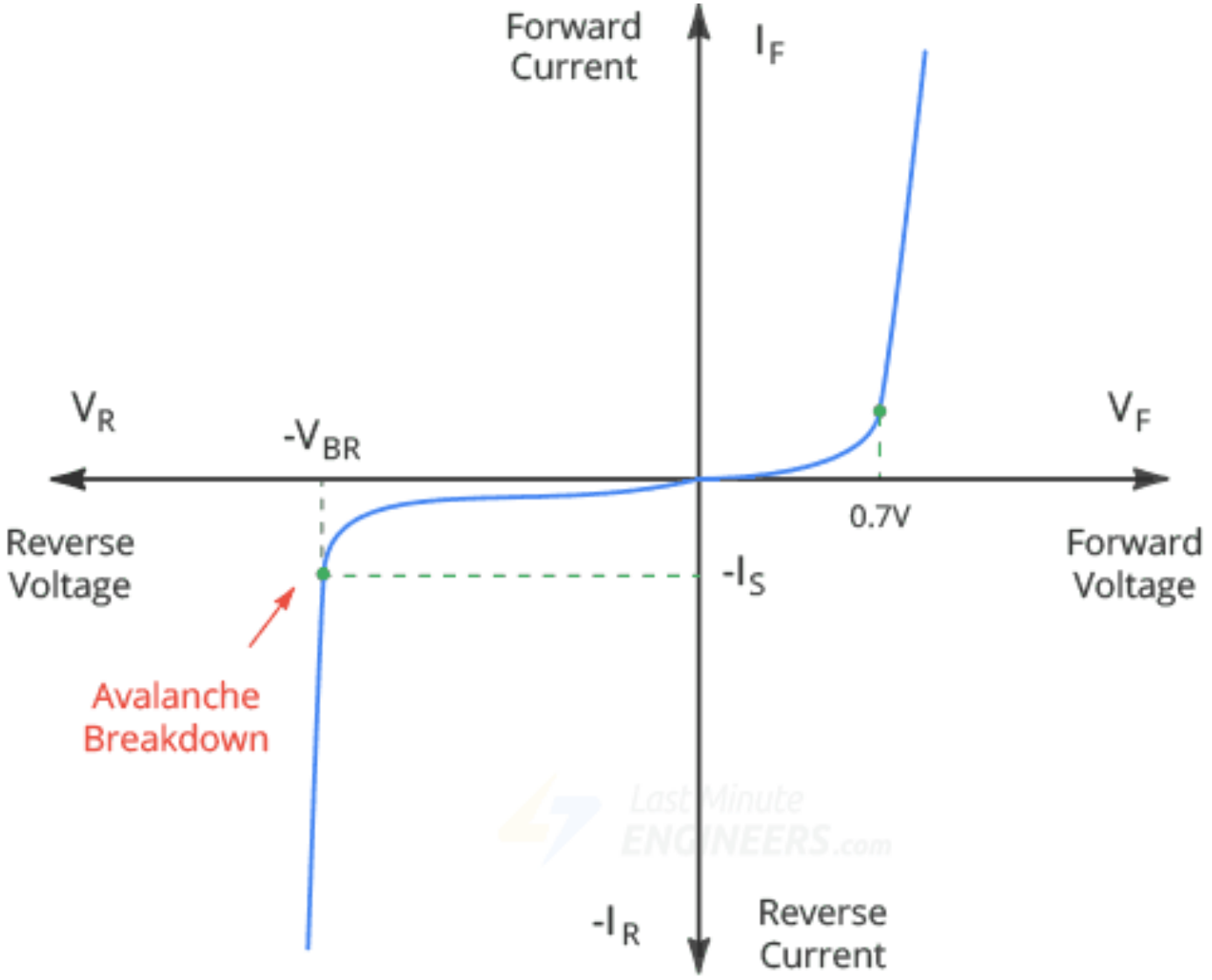


(i) Metallization

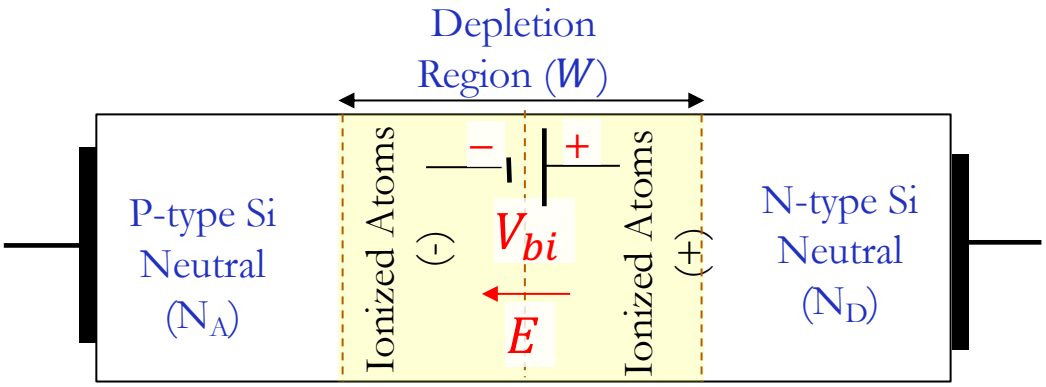


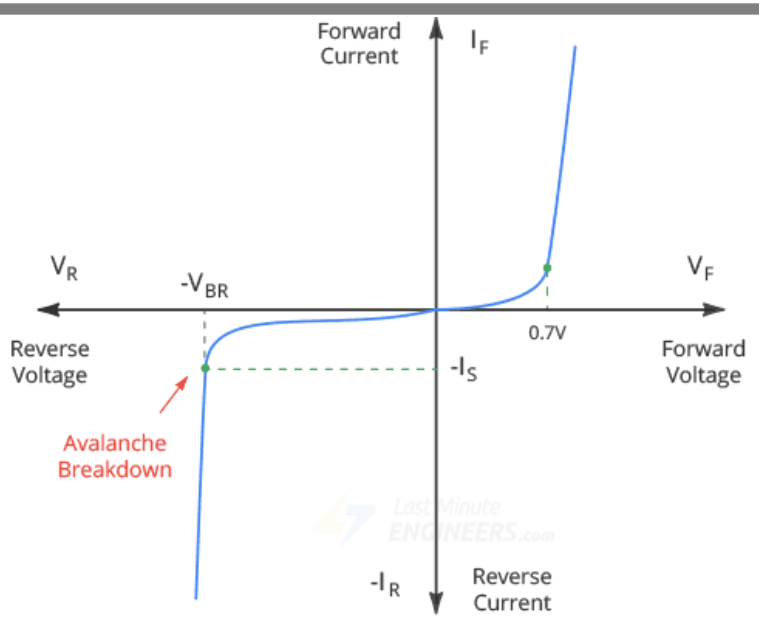
(j) Formed p-n junction

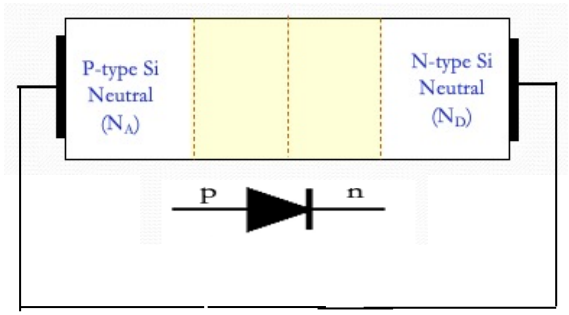
The PN junction: IV Curve



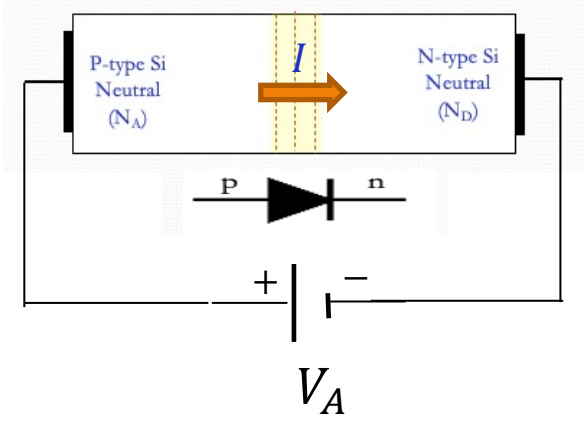
PN Junction Current and Biasing



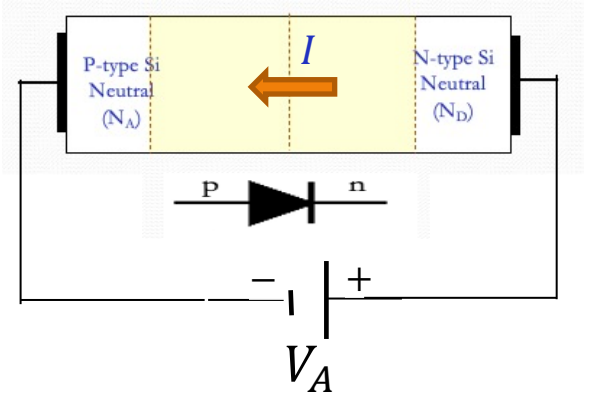




No Bias $I = 0$



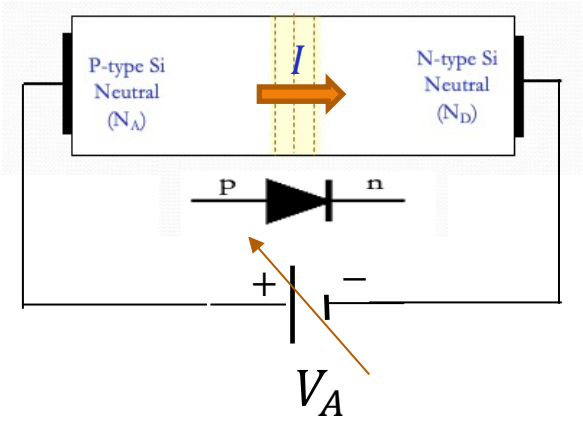
Forward Bias $I \neq 0$



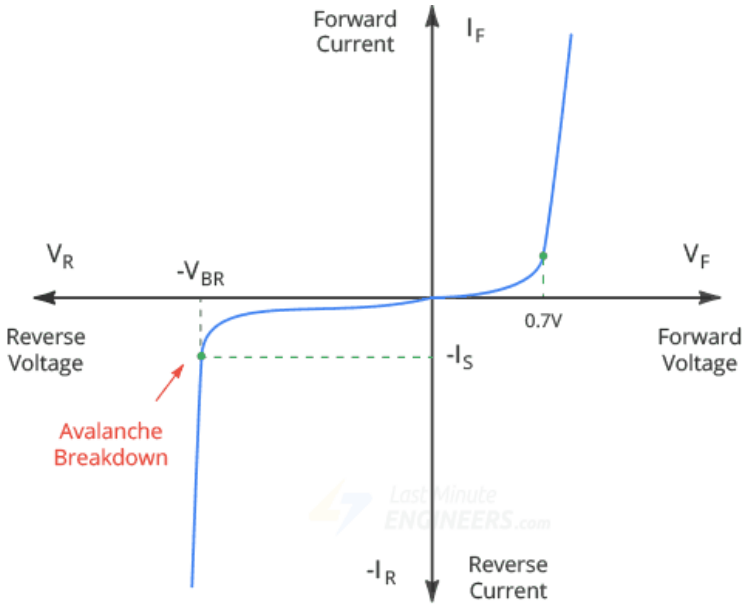
Reverse Bias $I = 0$

Breakdown $I \neq 0$

Current in Forward Region



Forward Bias $I \neq 0$



$$I = I_s\,(e^{V_D/nV_T} - 1)$$

- I: Current through junction (A)
- I_s: Saturation current (A)
- V_D: Voltage across junction (V)
- n: Ideality factor (1~2)
- V_T: Thermal voltage (V)

Q & A

Send me an email if
you have any
questions.